O ‘ QUVQO ‘ LLANMA · 2 0 2 6 - YILKUZ

Chiziqli algebra I

5–7 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridan barcha savollar va formulalar — geometriya va chiqarish, matritsa operatsiyalari va ajratishlar, transponirlash, almashtirishlar va vektorli fazolar — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil ishlash va Midterm imtihoniga tayyorgarlik uchun tuzilgan

Toshkent texnologiya universiteti · 2026-yil Kuz

Manbalar: kurs konspektlari, darslar, ISA, amaliy mashqlar; G. Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e)$

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1 Kurs xaritasi va hujjatdan qanday foydalanish

Hafta	Chiziqli algebra I
5	Geometriya va chiqarish: row picture va column picture, Gauss chiqarishi, teskari almashtirish, chiqarish matritsalari E, almashtirish matritsalari P, rang bilan bog‘lanish, $A = LU$ (Sträng §§2.1–2.3)
6	Matritsa amallari: ko‘paytirish qoidalari va AB ning to‘rt talqini, maxsus matritsalar, teskari matritsalar va ularning chiqarilishi, chiqarish matritsalari, LU/LDU-ajralishlari, transponirliklar, almashtirishlar (Sträng §§2.4–2.7)
7	Transponirliklar va almashtirishlar: transponirlik identifikatorlari, simmetrik matritsalar, 3×3 oltita almashtirish; vektorli fazolar va kichik fazolar, ustun fazosi $C(A)$, nol fazosi $N(A)$ (Sträng §§2.7, 3.1)

Ushbu qo‘llanma «Chiziqli algebra I — 1–4 haftalar» rasmiy to‘plamini davom ettiradi va 2026-yil kuz semestrining 5–7 haftalarini qamrab oladi. Unda har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar to‘plangan — har bir masala to‘liq bosqichma-bosqich yechim bilan, asosiy to‘plamdagi kabi.

Qo‘llanma qamrab olgan kurs fayllari (5–7 haftalar): ma’ruzalar; yechimlari bilan amaliy mashg‘ulotlar va dars masalalari; mustaqil ish topshiriqlari (ISA); 5-hafta kengaytirilgan Beamer nazariy konspekti; EXTRA material (6-hafta misollar bilan takrorlash). Darsliklar: G. Strang, Introduction to Linear Algebra (4-nashr) va parallel Linear Algebra and Its Applications (4-nashr).

Har bir hafta qanday tuzilgan. Avval — formulalar katalogi: haftaning barcha ta’riflari, teoremalari va qoidalari izohlar bilan. So‘ngra har bir manba fayli uchun «savollar va to‘liq yechimlar» bo‘limi: dars masalalari, amaliy mashg‘ulot, mustaqil ish topshiriqlari (ISA) va qo‘shimcha material. Masalalar so‘zma-so‘z keltirilgan; yechimlar to‘liq, bosqichma-bosqich, tekshiruvlar bilan.

Belgilash. Matritsalar qatorma-qator yoziladi: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — birinchi qator ab, ikkinchi qator cd; ustunli vektor $\begin{bmatrix} x \\ y \end{bmatrix}$, qatorli vektor $\begin{bmatrix} x & y \end{bmatrix}$ deb yoziladi. Darajalar va indekslar ^{} va _ orqali: A^{-1}, A^T, a_{11}. 5–7 haftalar kursning birinchi yarmidan hisoblanadi: chiqarish va matritsa amallari, $A = LU = LDU = PA$ ajratishlari, transponirlash, almashtirishlar va vektorli fazolar. 1–4 haftalar materiali bilan birga ular 8-hafta Midterm uchun ma’lum deb hisoblanadigan bazani tashkil qiladi.

2 5-hafta — Geometriya va chiqarish (Chiziqli algebra I)

[Manba: 5-hafta «Geometry and Elimination» ma’ruza konspekti (Chiziqli algebra $I, TT2026$): Strang, Introduction to Linear

Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, §§1.2–1.4]

2.6.1 Rang bilan bog‘liqlik: n ta asoschi elementlar teskari matritsa mavjudligini bildiradi

Takrorlash: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ rangi 1 va teskari matritsaga ega emas; $Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ cheksiz ko‘p

yechimga, $Ax = \begin{bmatrix}3\\7\end{bmatrix}$ esa hech qanday yechimga ega emas. Asosiy fakt: $n \times n$ kvadrat matritsa uchun n ta asoschi element (pivot) mavjudligi teskari matritsa mavjudligiga teng — bu holda $Ax = b$ barcha b uchun yagona yechimga ega; eliminatsiya topilgan asoschi elementlar soni rangdir. Ikkala algoritm ham $[A|b]$ da ishlaydi: Gauss–Jordan matritsani qisqartirilangan bosqichli shakl (reduced echelon form) yetkazadi; Gauss eliminatsiyasi yuqori uchburchak U da to‘xtaydi va teskari almashtirish bilan tugaydi.

$$A = \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix} : \text{rank}A = 1, A^{-1}yo'q;$$

$$Ax = \begin{bmatrix}3\\6\end{bmatrix} —cheksizko'pyechimlar,$$

$$Ax = \begin{bmatrix}3\\7\end{bmatrix} —bittahamyo'q$$

2.6.2 Qator tasviri: qatorlar to‘g‘ri chiziqlar va tekisliklar sifatida

$$2x - y = 1, x + y = 5;$$

$$\begin{bmatrix}2 & -1\\1 & 1\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$$

Har bir qator — bitta geometrik cheklov: 2D da to‘g‘ri chiziqlar, 3D da tekisliklar; yechim — ularning umumiy nuqtasi. Misol to‘g‘ri chiziqlari $(x, y) = (2, 3)$ da kesishadi; 3D da uchta tekislik:

$$\begin{bmatrix}2 & 1 & 1\\4 & -6 & 0\\-2 & 7 & 2\end{bmatrix} \cdot \begin{bmatrix}u\\v\\w\end{bmatrix} = \begin{bmatrix}5\\-2\\9\end{bmatrix} —tekisliklar(u,$$

$$v, w) = (1, 1, 2)dakesishadi$$

2.6.3 Ustun tasviri: Ax — ustunlarning chiziqli birikmasi

$$x \cdot \begin{bmatrix}2\\1\end{bmatrix} + y \cdot \begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix};$$

$$tekshirish:$$

$$2 \cdot \begin{bmatrix}2\\1\end{bmatrix} + 3 \cdot \begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$$

Strang ta’kidlaydi: Ax — A ustunlarining chiziqli birikmasidir — $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$ bo‘lishi uchun koeffitsientlarni tanlaymiz. Uch o‘lchovli variant (og‘irliklar (1, 1, 2)):

$$u \cdot \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \cdot \begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + w \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Ikkala tasvir ham bir xil $Ax = b$ tenglamaga: qatorlar — kesishishlar (tenglamalar), ustunlar — chiziqli birikmalar (vektorlar).

2.6.4 Uchta mumkin bo‘lgish natija

- Bitta yechim — mustaqil cheklovlar. • Cheksiz ko‘p yechimlar: $x + y = 2$ va 2x + 2y $= 4$ — ikkinchi tenglama

birinchisining ikki barobar katta, to‘g‘ri chiziqlar ustma-ust tushadi.

- Yechim yo‘q: $x + y = 2$ va 2x + 2y $= 5$ — chap tomonlar proporsional, o‘ng

tomonlar emas, to‘g‘ri chiziqlar parallel.

$$istisnochiqarishnatijaniko'rsatadi:$$
$$[00\ldots0|0] \to 0 = 0—cheksizko'pyechimlar; [00\ldots0|c], c$$

$$\neq 0, \to 0 = c—yechimyo'q$$

2.6.5 Asoschi element, ko‘paytuvchi va to‘g‘ridan-to‘g‘ri yurish

$$\lambda_{ij} = \frac{a_{ij}}{a_{jj}}, amaliyotR_i \leftarrow R_i - \lambda_{ij} \cdot R_j$$

a_{ij} elementini nol bo‘lmagan asoschi element a_{jj} ostidan chiqarish uchun (ko‘paytuvchi bu yerda λ_{ij} bilan belgilanadi), i-satrga j-satrni λ_{ij} ga ko‘paytirib ayiriladi. To‘liq misol: $\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]$; asoschi element 2, ko‘paytuvchilar $\lambda_{21} = \frac{4}{2} = 2va\lambda_{31} = -\frac{2}{2} = -1$:

$$R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 + R_1:$$
$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right]$$

Ikkinchi asoschi element −8, ko‘paytuvchi $\lambda_{32} = 8/(-8) = -1$, operatsiya $R_3 \leftarrow R_3 + R_2$. Uchburchak tizma va to‘g‘ridan-to‘g‘ri yurish natijasi:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right],$$

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix},$$

$$asosiyelementlar 2, -8, 1$$

2.6.6 Teskari almashtirish

$$w = 2; -8v - 2w = -12 \to v = 1; 2u + v + w = 5 \to u = 1;$$

$$x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Uchburchak tizma pastdan yuqoriga echiladi: oxirgi tenglamada faqat w, oldingisida $vvaw$ bor. Javobni asl tenglamaga almashtirib tekshiriladi: $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$. Satrni almashtirish teskari amalga ega (2R_1} ni qo‘shib qo‘ying), shuning uchun eliminatsiya yechimlarni saqlaydi; lekin operatsiya chap va o‘ng tomonlarga ham qo‘llaniladi, aks holda tenglamalar ekvivalent bo‘lmaydi.

2.6.7 Eliminatsiya E matritsalari

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$E_{21} \cdot A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix},$$

$$E_{21} \cdot b = \begin{bmatrix} 5 \\ -12 \\ 9 \end{bmatrix}$$

Chapdan ko‘paytirish qator operatsiyasini amalga oshiradi va o‘ng tomon ham xuddi shu matritsa bilan o‘zgaradi: $Ax = b (E_{21}A)x = E_{21}b$ ga aylanadi. U gacha to‘liq yurish (o‘ngdagi ko‘paytuvchi birinchi ishlaydi — ko‘paytirish tartibi muhim):

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$$E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$$

Bosqichni bekor qilish: ayirish o‘rniga qo‘shish kiritiladi — $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ uchun teskari $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} va E^{-1} \cdot E = I.$

2.6.8 Nol asoschi element: almashtirish P matritsalari

$$\left[\begin{array}{cc|c} 0 & 2 & 4 \\ 3 & 1 & 7 \end{array}\right] \to \left[\begin{array}{cc|c} 3 & 1 & 7 \\ 0 & 2 & 4 \end{array}\right] (qatorlarni almashtirish),$$

$$keyin y = 2, x = \frac{5}{3}$$

Nol asoschi element mutlaqo aynanlikni bildirmaydi: pastda nol bo‘lmagan element bo‘lsa, satrlar almashtiriladi. Almashtirish almashtirish matritsasi bilan amalga oshiriladi va u o‘ng tomonni ham o‘zgartiradi:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix},$$

$$ekvivalent sistema PAx = Pb$$

P xossalari: har bir qator va ustunda bitta 1, qolganlari nol; $P^{-1} = P^T$; bitta almashtirish uchun $P^2 = I$; $1 va 3$-qatorlarni almashtirish — $P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; chapdan ko‘paytirish tenglamalarni, emas, noma’lumlarni o‘zgartiradi. Eliminatsiyadagi muvaffaqiyatsizliklar: vaqtinchalik — nol asosiy elementda uning pastida nol bo‘lmagan element (almashtirish yordam beradi); doimiy — ustunda mavjud asosiy element yo‘q (singanlik: yechim yo‘q yoki cheksiz ko‘p yechim).

2.6.9 Oldinga qarash: $A = LU va PA = LU$

$$almashtirishsizA = LU(U—yuqoriuchburchak, Lchiqaruvchilarnisaqlaydi);$$
$$almashtirishlarbilan$$

$$PA = LU$$

Hafta yakuni: qatorlar — kesishishlar, ustunlar — kombinatsiyalar; eliminatsiya — asosiy elementlar ostidagi nollar va teskari almashtirish; E — qatorni almashtirish, P — qatorlarni almashtirish. LU-ajratish ($A = LU, PA = LU$)6-haftada quriladi.

[Manba: 5-hafta dars masalalari (Proyeksiya), masala $1 - 15(law5classg01 - g11va23 - 31$-bet matn qatlami)]

Mashq 1. Qatorlar va ustunlar tasviri. Tizimni $\begin{bmatrix}2 & -1\\ 1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}1\\ 5\end{bmatrix}$ yeching va ikkala geometrik tasvirlarni tavsiflang.

Yechim. Tenglamalarni qo‘shamiz: 3x $= 6, x = 2$, undan $y = 5-2 = 3$. 2x $-y = 1vax+y =$ 5 to‘g‘ri chiziqlari (2, 3) nuqtada kesishadi; ustunlar: $2\cdot\begin{bmatrix}2\\ 1\end{bmatrix} + 3\cdot\begin{bmatrix}-1\\ 1\end{bmatrix} = \begin{bmatrix}4 & & 3\\ 2 & + & 3\end{bmatrix} = \begin{bmatrix}1\\ 5\end{bmatrix}$. Javob: $x = 2, y = 3$.

Mashq 2. Uch holat. Tizimni $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}\cdot\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}2\\ c\end{bmatrix}$ $c = 4vac = 5$ uchun tasniflang.

Yechim. $R_2 \leftarrow R_2 - 2R_1[00|c-4]$ qatorni beradi. $c = 4dabu0 = 0$: $x + y = 2$ qoladi, cheksiz ko‘p yechimlar — $\begin{bmatrix}2 & t\\ t & \end{bmatrix}$. $c = 5dabu0 = 1$ — ziddiyat, yechim yo‘q.

Mashq 3. Ko‘paytchini toping. Kengaytirilgan matritsada $\left[\begin{array}{cc|c}2 & 3 & 7\\ 8 & 13 & 29\end{array}\right]$ birinchi asosiy element ostidagi hadni chiqaring va ikkala asosiy elementni toping.

Yechim. Ko‘paytchi $\lambda_{21} = \frac{8}{2} = 4$. $R_2 \leftarrow R_2 - 4R_1$: $[8,\ 13|29] - 4\cdot[2,\ 3|7] = [0,\ 1|1]$. Natija: $\left[\begin{array}{cc|c}2 & 3 & 7\\ 0 & 1 & 1\end{array}\right]$; asosiy elementlar $2va1$.

Mashq 4. Orqaga almashtirish. Tizimni $\begin{bmatrix}2 & 1 & -1\\ 0 & 3 & 2\\ 0 & 0 & 4\end{bmatrix}\cdot\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}3\\ 7\\ 8\end{bmatrix}$ yeching.

Yechim. Pastdan yuqoriga: 4z $= 8, z = 2$; 3y $+2\cdot 2 = 7, y = 1$; 2x $+1 - 2 = 3, x = 2$. Javob: $x = \begin{bmatrix}2\\ 1\\ 2\end{bmatrix}$. Tekshirish: $2\cdot 2 + 1 - 2 = 3,\ 3 + 4 = 7,\ 8 = 8$ — to‘g‘ri.

Mashq 5. To‘liq chiqarib tashlash. Tizimni

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

chiqarib tashlash usuli bilan yeching.

Yechim. $R_2 \leftarrow R_2 - 2R_1 \to [01-1|-2]$; $R_3 \leftarrow R_3 - R_1 \to [012|8]$; $R_3 \leftarrow R_3 - R_2 \to [003|10]$, ya’ni 3z $= 10$. Orqaga almashtirish: $z = \frac{10}{3}$; $y - z = -2 \to y = \frac{4}{3}$; $x + y + z = 6 \to x = \frac{4}{3}$. Javob: [4/3; 4/3; 10/3]. Tekshirish: qatorlar 6, 10, 14 beradi — to‘g‘ri.

Mashq 6. Elementar matritsani aniqlang. 3×3 matritsida $R_3 \leftarrow R_3 - 4R_1$ amalini amalga oshiruvchi E ni toping.

Yechim. E — birlik matritsida (3, 1) o‘rinda -4 bo‘lgan matritsa: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; EA ning uchinchi qatori (3-qator) $-\,4\cdot$(1-qator), 1–2-qatorlar o‘zgarmaydi. Aks operatsiya — qo‘shish: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$ $da(3,\,1)$ o‘rinda $+4$.

Mashq 7. EA ko‘paytmasi. $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ $vaA = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ bo‘lsin. EA ni hisoblang va amalni tushuntiring.

Yechim. Ko‘paytmaning 1-qatori: $\begin{bmatrix} 3 & 1 \end{bmatrix}$; 2-qatori: $-2\cdot\begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 3 \end{bmatrix}$. Shunday qilib, $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$: $E2$-qatordan 1-qatarning ikkilanganini ayiradi — $\frac{6}{3} = 2$ ko‘paytchili chiqarib tashlash.

Mashq 8. O‘zgartirilgan o‘ng tomon. $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ $vab = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$ uchun $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ dan foydalanib, tizimni yeching.

Yechim. $Eb = \begin{bmatrix} 4 \\ -8 \;+\; 17 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; mashq 7 dagi $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ bilan birga 3y $= 9, y = 3$ va 3x $+3 = 4, x = \frac{1}{3}$. Asl tizimda tekshirish: $3\cdot(\frac{1}{3}) + 3 = 4$, $6\cdot(\frac{1}{3}) + 5\cdot 3 = 17$ — to‘g‘ri.

Mashq 9. Nol asosiy element. Tizimni $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}\cdot\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ qatorlarni almashtirib yeching; P ni toping.

Yechim. (1, 1) o‘rinda nol — vaqtacha muvaffaqiyatsizlik: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Unda 2y $= 4, y = 2$; 3x $+y = 7, x = \frac{5}{3}$. Tekshirish: $2\cdot 2 = 4$, $3\cdot(\frac{5}{3}) + 2 = 7$ — to‘g‘ri; nol asosiy element aynanlikni anglatmadi.

Mashq 10. Uch qatorni almashtirish. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ **matritsasining** $1va3$**-qatorlarini almashtiruvchi** P **ni toping;** PA **ni hisoblang.**

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — birlik matritsining qatorlari 3, 2, 1 tartibida; PA ning 1-qatori A matritsasining 3-qatori va $h.k.$: $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. $1 \leftrightarrow 3$ almashtirishi — involutsiya: $P^{-1} = PvaP^2 = I (P^{-1} = P^T$ bilan mos keladi).

Mashq 11. Aynan, lekin mos keladigan. Tizimni $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ **yeching.**

Yechim. 2-qatori o‘ng tomoni $(6 = 2 \cdot 3)1$-qatordan ikki barobar katta, shuning uchun $R_2 \leftarrow R_2 - 2R_1 0 = 0$ beradi. $x+$ 2y $+z = 3vay + z = 2$ qoladi; erkin o‘zgaruvchi $z = t$: $y = 2 - t$, $x + 2(2 - t) + t = 3 \to x = t - 1$. Javob: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$ — cheksiz ko‘p yechimlar. $t = 1$ da tekshirish: $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ $3, 6, 2$ beradi — to‘g‘ri.

Mashq 12. Aynan va mos kelmaydigan. Oldingi tizimda o‘ng tomonni $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$ **ga o‘zgartiring. Tizim mos keladimi?**

Yechim. Endi $7 \neq 2 \cdot 3$. $R_2 \leftarrow R_2 - 2R_1 [000|7 - 6] = [000|1]$ qatorni beradi, ya’ni $0 = 1$ — ziddiyat. Tizim mos kelmaydi: o‘ng tomon zarur shart $b_2 = 2b_1$ ni buzadi.

Mashq 13. Ustunli mos kelish talqini. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ **uchun qaysi** $b = [b_1$**;** $b_2]$ **vektorlari** $Ax = b$ **tenglamani qondiradi?**

Yechim. Ikkinchi qator birinchisining ikki barobar katta: $R_2 \leftarrow R_2 - 2R_1$ amalidan keyin $[00|b_2 - 2b_1]$ qoladi, shuning uchun moslik $b_2 = 2b_1$ ni talab qiladi. $\begin{bmatrix} 1 \\ 2 \end{bmatrix} va \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ustunlari $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ga karrali, shuning uchun Ax har doim $b_2 = 2b_1$ to‘g‘ri chiziqda yotadi. Javob: $b = [b_1;\ 2b_1]$ — b ustunlar A orqali o‘tadigan to‘g‘ri chiziqda yotishi kerak.

Mashq 14. Matritsa amallari tartibi. $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} vaP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **deb belgilaymiz.** $EPvaPE$ **ni solishtiring.**

Yechim. EP: 1-qator $= \begin{bmatrix}0 & 1\end{bmatrix}$, 2-qator $= -2 \cdot \begin{bmatrix}0 & 1\end{bmatrix} + \begin{bmatrix}1 & 0\end{bmatrix} = [1, -2]$, ya'ni $EP = \begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}$. PE: 1-qator $= 1 \cdot \begin{bmatrix}-2 & 1\end{bmatrix} = \begin{bmatrix}-2 & 1\end{bmatrix}$, 2-qator $= \begin{bmatrix}1 & 0\end{bmatrix}$, ya'ni $PE = \begin{bmatrix}-2 & 1\\ 1 & 0\end{bmatrix}$. Ko'paytmalar har xil: qator amallari, umuman olganda, kommutativ emas.

Mashq 15. Murojaat: tizim qurish. Qatorlarni almashtirishni talab qiladigan, uchta asosiy elementga va $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$ yechimga ega bo'lgan 3×3 tizimini qurib ko'ring.

Yechim. (1, 1) o'rningidagi nol va undan pastdagi nol bo'lmagan element: $A = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$; o'ng tomonni ustun tasviri bilan belgilaymiz: $b = A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$. $1 va 2$-qatorlarni almashtirish ($P = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$)$A$ ni birlik matritsiga aylantiradi: asosiy elementlar 1, 1, 1, yechim $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Tekshirish: $A \cdot \begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$ — to'g'ri.

[Manba: 5-hafta amaliy sessiya masalalari yechimlari (Week$5TA$ Problems Projection), masalalar 1–15; matritsat satrli matn
qatlamidan tiklangan]

Mashq 1. Ikkita to'g'ri chiziq sistemasini yeching $\begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}7\\8\end{bmatrix}$.

Yechim. $R_2 \leftarrow R_2 - 3R_1$: $(1-6)y = 8-21$, ya'ni $-5y = -13, y = \frac{13}{5}$; undan $x = 7 - 2\cdot(\frac{13}{5}) = \frac{9}{5}$. Javob: (9/5, 13/5). Tekshirish: $\frac{9}{5} + \frac{26}{5} = 7$, $\frac{27}{5} + \frac{13}{5} = 8$ — to'g'ri.

Mashq 2. Parallel yoki ustma-ust tushadimi? Sistemaning $\begin{bmatrix}2 & 4\\ 1 & 2\end{bmatrix} \cdot \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}6\\c\end{bmatrix}$ $nic = 3 va c = 4$ uchun tasniflang.

Yechim. 1-satr 2-satrdan ikki marta katta chap tomonda; moslik o'ng tomonda ham xuddi shuni talab qiladi: $6 = 2c$, ya'ni $c = 3$. $c = 3$ bo'lganda, $x + 2y = 3$ — cheksiz ko'p yechimlar $\begin{bmatrix}3 - 2t\\ t\end{bmatrix}$ (to'g'ri chiziqlar ustma-ust tushadi). $c = 4$ bo'lganda, ikkinchi tenglamaning ikkilanganidan $2x + 4y = 8 \neq 6$ — yechim yo'q (to'g'ri chiziqlar parallel).

Mashq 3. Bitta eliminatsiya bosqichi. Matritsadagi $\begin{bmatrix}3 & 2\\ 9 & 8\end{bmatrix}$ birinchi asosiy element ostidagini yo'qing.

Yechim. $\lambda_{21} = \frac{9}{3} = 3$ ko'paytuvchisi; $R_2 \leftarrow R_2 - 3R_1$: $[9-9, 8-6] = [0, 2]$. Natija: $\begin{bmatrix}3 & 2\\ 0 & 2\end{bmatrix}$; asosiy elementlar $3 va 2$.

Mashq 4. Uchburchakli tizim. $\begin{bmatrix}1&2&1\\0&2&-2\\0&0&3\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}8\\0\\6\end{bmatrix}$ **tizimini yeching.**

Yechim. Pastdan yuqoriga: 3z $=6, z=2$; 2y $-2\cdot 2=0, y=2$; $x+4+2=8, x=2$. Javob: $\begin{bmatrix}2\\2\\2\end{bmatrix}$. Tekshirish: $2+4+2=8$, $4-4=0$, $3\cdot 2=6$ — to‘g‘ri.

Mashq 5. Uch noma’lumli to‘liq tizim.

$$\left[\begin{array}{ccc|c}1&1&0&3\\2&3&1&9\\1&0&1&4\end{array}\right]$$

tizimini yeching.

Yechim. $R_2\leftarrow R_2-2R_1\to[011|3]$; $R_3\leftarrow R_3-R_1\to[0-11|1]$; $R_3\leftarrow R_3+R_2\to[002|4]$, ya’ni 2z $=4, z=2$. Orqaga: $y+z=3\to y=1$; $x+y=3\to x=2$. Javob: $\begin{bmatrix}2\\1\\2\end{bmatrix}$. Tekshirish: $2+1=3$, $4+3+2=9$, $2+2=4$ — to‘g‘ri.

Mashq 6. E_{32} **ni tuzing.** $R_3\leftarrow R_3+5R_2$ **amali uchun** 3×3 **matritsani yozing.**

Yechim. E_{32} — bu (3, 2) o‘rinda 5 bo‘lgan birlik matritsa: $E_{32}=\begin{bmatrix}1&0&0\\0&1&0\\0&5&1\end{bmatrix}$; 1–2-qatorlar o‘zgarmaydi, 3-qator ko‘paytmasi esa (3-qator) $+\,5\cdot$ (2-qator) ga teng.

Mashq 7. Amalni teskarisiga o‘tkazish. $E=\begin{bmatrix}1&0&0\\0&1&0\\0&-3&1\end{bmatrix}$ **matritsasi uchun teskarisini toping.**

Yechim. $ER_3\leftarrow R_3-3R_2$(2−qatorning uch baravarini ayirish) amalini amalga oshiradi; teskari amal — qo‘shish: $E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\0&3&1\end{bmatrix}$ (belgisi o‘zgardi). Tekshirish: $E^{-1}\cdot E=I$.

Mashq 8. O‘ng tomonni o‘zgartirish. $E=\begin{bmatrix}1&0&0\\-2&1&0\\0&0&1\end{bmatrix}$ $nib=\begin{bmatrix}3\\5\\7\end{bmatrix}$ **ga qo‘llang.**

Yechim. $Eb=[3;\ 5-2\cdot 3;\ 7]=\begin{bmatrix}3\\-1\\7\end{bmatrix}$: ikkinchi komponent $5-2\cdot 3=-1$ bo‘ladi. Ex ni ikkinchi tenglamadan 2 koeffitsient bilan chiqarib tashlaydi — o‘ng tomon bir xil amal bilan o‘zgaradi.

Mashq 9. Qatorlarni almashtiruvchi matritsa. 3×3 **matritsaning** $2va3$**-qatorlarini o‘rnini almashtiruvchi** P **ni toping.**

Yechim. $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ — bu $2va3$-qatorlari almashtirilgan birlik matritsa. Ikkita almashtirish boshlang‘ich tartibni qaytaradi: $P^2 = I$, shuning uchun $P^{-1} = P(P^{-1} = P^T$ bilan mos keladi); chapdan ko‘paytirish tenglamalarni, noma’lumlarni emas almashtiradi.

Mashq 10. Almashtirishni talab qiladigan tizim. $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$ **tizimini yeching.**

Yechim. (1, 1) o‘rnidagi asosiy element nol — satrlarni almashtiramiz: 2x + 3y = $14vay = 4$. Unda 2x $+12 = 14, x = 1$. Javob: (1, 4). Tekshirish: $0 \cdot 1 + 4 = 4, 2 + 12 = 14$ — to‘g‘ri.

Mashq 11. Erkin o‘zgaruvchi. Tizimni yeching $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$**.**

Yechim. Uchinchi satr — $0 = 0$, cheklov bermaydi; asosiy elementlar ikkita, noma’lumlar uchta. $z = t$ deb olaylik: $y + t = 3 \to y = 3 - t$; $x + (3 - t) + t = 4 \to x = 1$. Javob: $\begin{bmatrix} 1 \\ 3 \quad t \\ t \end{bmatrix}$. $t = 0$ da tekshirish: $\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ $4, 3, 0$ beradi — to‘g‘ri.

Mashq 12. Ziddiyatli satr. Tushuntiring, nega

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

tizimining yechimi yo‘q.

Yechim. Oxirgi satr $0 \cdot x + 0 \cdot y + 0 \cdot z = 1$ deb o‘qiladi, ya’ni $0 = 1$ — imkonsiz tenglik: chap tomondagi nol satr hech qanday x, y, z uchun o‘ng tomondagi nol bo‘lmagan qiymatni bera olmaydi. Tizim ziddiyatli.

Mashq 13. $EPvaPE$ **ni solishtiring.** $E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} vaP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **uchun ikkala ko‘paytmani hisoblang.**

Yechim. EP: 1-satr $= \begin{bmatrix} 0 & 1 \end{bmatrix}$, 2-satr $= 4 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} = [1, 4]$, ya’ni $EP = \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$. PE: 1-satr $= 1 \cdot \begin{bmatrix} 4 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \end{bmatrix}$, 2-satr $= \begin{bmatrix} 1 & 0 \end{bmatrix}$, ya’ni $PE = \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$. Ko‘paytmalar har xil: «avval chiqarib tashlash, keyin almashtirish» va «avval almashtirish, keyin chiqarib tashlash» turli matritsalar beradi.

Mashq 14. Ustun fazosini tekshirish. $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ **tizimini yechish mumkinmi?**

Yechim. Ikkala ustun ham ikkinchi komponenti birinchisining ikki barobariga ega: $\begin{bmatrix}1\\2\end{bmatrix} va \begin{bmatrix}3\\6\end{bmatrix} = 3\cdot\begin{bmatrix}1\\2\end{bmatrix}$; $x\cdot\begin{bmatrix}1\\2\end{bmatrix} + y\cdot\begin{bmatrix}3\\6\end{bmatrix}$ har qanday kombinatsiyasi $\begin{bmatrix}s\\2s\end{bmatrix}$ ko‘rinishida bo‘ladi. $b = \begin{bmatrix}2\\5\end{bmatrix}$ uchun $5 \neq 2\cdot 2 = 4$ — b ustunlar fazosida yotmaydi. Javob: yo‘q (birinchi tenglamaning ikkilangan versiyasi 2x + 6y $= 4 \neq 5$ — to‘g‘ri chiziqlar parallel).

Mashq 15. Muhim: almashtirish matritsasini tuzish. $(R_1,\ R_2,\ R_3)$ **satrlarini** $(R_3,\ R_1,\ R_2)$ **ga olib boradigan** P **ni tuzing.**

Yechim. Yangi 1-satr — eski R_3, yangi 2 — eski R_1, yangi 3 — eski R_2, shuning uchun P satrlari birlik matritsasi satrlari 3, 1, 2 tartibda: $P = \begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$. $\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$ satrlarida tekshirish: $P\cdot A = \begin{bmatrix}7&8&9\\1&2&3\\4&5&6\end{bmatrix}$. Chapdan ko‘paytirish satrlarni almashtiradi; $P^3 = I$.

[Manba: 5-hafta Mustaqil o‘quv topshiriqlari, mashqlar 1–15; yechimlar faylda yo‘q — pastda to‘liq yechimlar tekshiruv bilan
keltirilgan]

Mashq 1. Satr va ustun tasviri. $\begin{bmatrix}2&-1\\1&1\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$ **tizimini yeching. Ikkala geometrik tasviri ham tavsiflang. Har bir chiqarib tashlash bosqichini va javobni tekshiring.**

Yechim. Tenglamalarni qo‘shamiz: 3x $= 6, x = 2, y = 3$. Tekshirish: $2\cdot 2 - 3 = 1, 2 + 3 = 5$. 2x $- y = 1 va x + y = 5$ to‘g‘ri chiziqlari (2, 3) da kesishadi; ustunlar: $2\cdot\begin{bmatrix}2\\1\end{bmatrix} + 3\cdot\begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$. Javob: $x = 2, y = 3$.

Mashq 2. Uchta natija. $\begin{bmatrix}1&1\\2&2\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}2\\c\end{bmatrix}$ **tizimini** $c = 4 va c = 5$ **uchun tasniflang. Har bir chiqarib tashlash bosqichini va javobni tekshiring.**

Yechim. $R_2 \leftarrow R_2 - 2R_1 [00|c - 4]$ beradi. $c = 4$ da: $0 = 0$, yechimlar $\begin{bmatrix}2&t\\t&\end{bmatrix}$ (tekshirish: $(2 - t) + t = 2, 2(2 - t) +$ 2t $= 4$). $c = 5$ da: $0 = 1$ — yechim yo‘q.

Mashq 3. Ko‘paytchini toping. $\left[\begin{array}{cc|c}2&3&7\\8&13&29\end{array}\right]$ **da birinchi asosiy element ostidagini chiqarib tashlang; ikkala asosiy elementni toping. Har bir bosqichni va javobni tekshiring.**

Yechim. $\lambda_{21} = \frac{8}{2} = 4$; $R_2 \leftarrow R_2 - 4R_1$: $[8,\ 13|29] - 4\cdot[2,\ 3|7] = [0,\ 1|1]$. Natija: $\left[\begin{array}{cc|c}2&3&7\\0&1&1\end{array}\right]$, asosiy elementlar $2 va 1$. Aylantirilishi bilan tekshirish: $[0,\ 1|1] + 4\cdot[2,\ 3|7] = [8,\ 13|29]$.

Mashq 4. Orqaga qaytish almashtirish. $\begin{bmatrix}2&1&-1\\0&3&2\\0&0&4\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$ **tizimini yeching. Har bir bosqichni va javobni tekshiring.**

Yechim. Pastdan yuqoriga: $z=\frac{8}{4}=2$; 3y $+2\cdot2=7, y=1$; 2x $+1-2=3, x=2$. Tekshirish: $\begin{bmatrix}2&1&-1\\0&3&2\\0&0&4\end{bmatrix}\cdot\begin{bmatrix}2\\1\\2\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$. Javob: $x=\begin{bmatrix}2\\1\\2\end{bmatrix}$.

Mashq 5. To‘liq chiqarib tashlash.

$$\left[\begin{array}{ccc|c}1&1&1&6\\2&3&1&10\\1&2&3&14\end{array}\right]$$

tizimini chiqarib tashlash usuli bilan yeching. Har bir bosqichni va javobni tekshiring.

Yechim. $R_2\leftarrow R_2-2R_1 va R_3\leftarrow R_3-R_1$ $\left[\begin{array}{ccc|c}1&1&1&6\\0&1&-1&-2\\0&1&2&8\end{array}\right]$ beradi; keyin $R_3\leftarrow R_3-R_2[003|10]$ beradi. Orqaga: $z=\frac{10}{3}$, $y=z-2=\frac{4}{3}$, $x=6-y-z=\frac{4}{3}$. Tekshirish: satrlar 6, 10, 14 beradi — to‘g‘ri. Javob: [4/3; 4/3; 10/3].

Mashq 6. Elementar matritsani aniqlang. 3×3 **matritsida** $R_3\leftarrow R_3-4R_1$ **amalga oshiradigan** E **ni toping. Har bir bosqichni va javobni tekshiring.**

Yechim. $E=\begin{bmatrix}1&0&0\\0&1&0\\-4&0&1\end{bmatrix}$ — birlik matritsa -4 (3, 1) o‘rnida; EA ning uchinchi qatori (qator $3)-4\cdot$(qator 1). Tekshirish $\begin{bmatrix}1&1&1\\2&3&1\\1&2&3\end{bmatrix}$ uchun: qator $3=\begin{bmatrix}1&2&3\end{bmatrix}-4\cdot\begin{bmatrix}1&1&1\end{bmatrix}=\begin{bmatrix}-3&-2&-1\end{bmatrix}$. Bekor qilish: $E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\4&0&1\end{bmatrix}$.

Amaliyot 7. EA **ko‘paytmasi.** $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$, $A=\begin{bmatrix}3&1\\6&5\end{bmatrix}$ **bo‘lsin.** EA **ni hisoblang va amalni tushuntiring. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. Qator 1: $\begin{bmatrix}3&1\end{bmatrix}$; qator 2: $\begin{bmatrix}6&5\end{bmatrix}-2\cdot\begin{bmatrix}3&1\end{bmatrix}=\begin{bmatrix}0&3\end{bmatrix}$; $EA=\begin{bmatrix}3&1\\0&3\end{bmatrix}$ — 2-qatordan 1-qatorning ikkilanganini ayirish (ko‘paytuvchi 2). Tekshirish: $E^{-1}=\begin{bmatrix}1&0\\2&1\end{bmatrix}A$ ni qaytaradi.

Amaliyot 8. O‘zgartirilgan o‘ng tomon. $A=\begin{bmatrix}3&1\\6&5\end{bmatrix} vab=\begin{bmatrix}4\\17\end{bmatrix}$ **uchun** $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}$

dan foydalanib sistemani yeching. Har bir bosqichni ko‘rsating va javobni tekshiring.

Yechim. $Eb = [4;\ -2\cdot 4+17] = \begin{bmatrix}4\\9\end{bmatrix}$; $EA = \begin{bmatrix}3 & 1\\0 & 3\end{bmatrix}$; keyin 3y $= 9 \to y = 3$ va 3x $+3 = 4 \to x = \frac{1}{3}$. Asl sistemada tekshirish: $3\cdot(\frac{1}{3})+3 = 4$, $6\cdot(\frac{1}{3})+5\cdot 3 = 17$ — to‘g‘ri.

Amaliyot 9. Nol asosiy element. $\begin{bmatrix}0 & 2\\3 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}4\\7\end{bmatrix}$ **ni qatorlarni almashtirib yeching;** P **ni toping. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $P = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$; $PA = \begin{bmatrix}3 & 1\\0 & 2\end{bmatrix}$, $Pb = \begin{bmatrix}7\\4\end{bmatrix}$; 2y $= 4 \to y = 2$, 3x $+2 = 7 \to x = \frac{5}{3}$. Tekshirish: $0\cdot(\frac{5}{3})+2\cdot 2 = 4$, $3\cdot(\frac{5}{3})+2 = 7$ — to‘g‘ri.

Amaliyot 10. Uchta qatorning almashtirilishi. $A = \begin{bmatrix}1 & 2 & 0\\0 & 1 & 3\\4 & 0 & 2\end{bmatrix}$ **matritsasi** $1 va 3$**-qatorlarini almashtiruvchi** P **ni toping;** PA **ni hisoblang. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $P = \begin{bmatrix}0 & 0 & 1\\0 & 1 & 0\\1 & 0 & 0\end{bmatrix}$; PA ning 1-qatori A ning 3-qatori va PA ning 3-qatori A ning 1-qatori, shuning uchun $PA = \begin{bmatrix}4 & 0 & 2\\0 & 1 & 3\\1 & 2 & 0\end{bmatrix}$. Tekshirish: $P^2 = I$, ya’ni $P\cdot PA = A$; $P^{-1} = P = P^T$.

Amaliyot 11. Ayniy, lekin mos keladigan. $\begin{bmatrix}1 & 2 & 1\\2 & 4 & 2\\0 & 1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}3\\6\\2\end{bmatrix}$ **ni yeching. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $R_2 \leftarrow R_2 - 2R_1[000|6-2\cdot 3] = [000|0]$ — qator yo‘qoldi, sistema mos, lekin ayniy. Qoladi $x+$ 2y $+z = 3 va y + z = 2$; $z = t$: $y = 2-t, x = t-1$. Javob: $\begin{bmatrix}t & 1\\2 & t\\t & \end{bmatrix}$. $t = 2$ da tekshirish: $\begin{bmatrix}1\\0\\2\end{bmatrix}$ 3, 6, 2 beradi — to‘g‘ri.

Amaliyot 12. Ayniy va mos kelmaydigan. Oldingi sistemada o‘ng tomonni $b = \begin{bmatrix}3\\7\\2\end{bmatrix}$ **ga o‘zgartiring. Sistema mos keladimi? Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $R_2 \leftarrow R_2 - 2R_1[000|7-2\cdot 3] = [000|1]$, ya’ni $0 = 1$ — ziddiyat. Sistema mos kelmaydi: zarur shart $b_2 = 2b_1 (7 \neq 6)$ buzilgan.

Amaliyot 13. Ustunlarning mos kelish talqini. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ **uchun qaysi** $b = [b_1; b_2]$ **vektorlari** $Ax = b$ **tenglamani yechimga ega? Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $R_2 \leftarrow R_2 - 2R_1 [12|b_1;\ 00|b_2 - 2b_1]$ ni qoldiradi; mos kelish uchun $b_2 = 2b_1$ talab qilinadi. Ustunlar $\begin{bmatrix} 1 \\ 2 \end{bmatrix} va \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ga karrali, shuning uchun Ax har doim $b_2 = 2b_1$ to‘g‘risida yotadi. Javob: $b = [b_1; 2b_1]$. Tekshirish: $\begin{bmatrix} 4 \\ 8 \end{bmatrix}$ $(x = 4, y = 0)$ yechimga ega, $\begin{bmatrix} 4 \\ 9 \end{bmatrix}$ esa yo‘q.

Amaliyot 14. Matritsa amallar tartibi. $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} vaP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **bo‘lsin.** $EPvaPE$ **ni solishtiring. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. EP: qatorlar $\begin{bmatrix} 0 & 1 \end{bmatrix} va [1,\ -2]$, ya’ni $EP = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$. PE: qatorlar $\begin{bmatrix} -2 & 1 \end{bmatrix} va \begin{bmatrix} 1 & 0 \end{bmatrix}$, ya’ni $PE = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. Ko‘paytmalar farq qiladi — qator amallari kommutativ emas: EP da chiqarish almashtirilgan qatorlarga ta’sir qiladi, $PEdaesa$ asl qatorlarga.

Amaliyot 15. Murojaat: sistemani qurish. Qatorlarni almashtirishni talab qiladigan, uchta asosiy elementga ega va $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ **yechimga ega bo‘lgan** 3×3 **sistemani qurib ko‘ring. Har bir bosqichni ko‘rsating va javobni tekshiring.**

Yechim. $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ — (1, 1) o‘rnida nol, pastda birlik; $b = A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$. $1va2$-qatorlarni almashtirish $(P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix})$ birlik matritsani beradi: asosiy elementlar 1, 1, 1, yechim $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Tekshirish: $A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$; almashtirmasdan birinchi ustun chiqarilmaydi.

3 6-hafta — Matritsa amallari, teskari matritsalar va ajratishlar (Chiziqli algebra I)

[Manba: 6-hafta darsi prezentatsiyasi (Strang, Introduction to Linear Algebra, 4th ed., §§2.4–2.7; qo‘shimcha: Lay, §§1.4–1.6)]

3.5.1 5-haftani takrorlash: sistemaning ikki tasviri va chiqarish

Satr tasviri (row picture): tenglamalar — kesishgan to‘g‘ri chiziqlar yoki tekisliklar. Ustun tasviri (column picture): $Ax = x_1 \cdot a_1 + ... + x_n \cdot a_n = b$. Misol: $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ $(x,\ y) = (2,\ 1)$ ni beradi; ikkala tasvir ham bir sistemani tasvirlaydi. Chiqarish: asosiy element (pivote)$a_{kk} \neq 0$ bo‘lganda, $m_{ik} = a_{ik}/a_{kk}$ ko‘paytuvchi olinadi va $R_i \leftarrow R_i - m_{ik} \cdot R_k$ amalga oshiriladi; shunday $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, keyin teskari almashtirib uchburchak sistemani yechamiz. Aynanlik: $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ cheksiz ko‘p yechimga ega, $6ni7$ ga almashtirish $0 = 1$ ziddiyatini beradi. Kvadrat matritsalar uchun to‘liq asosiy elementlar to‘plami har qanday o‘ng tomondagi uchun yagona yechimni anglatadi.

3.5.2 Ko‘paytirish qoidalari va AB ning to‘rt talqini

$$A(\text{m×n}), B(\text{n×p}) \rightarrow AB(\text{m×p}), (AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$$

Misol: $\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 4 \end{bmatrix}$. AB ga to‘rtta qarash: (1) element bo‘yicha — A matritsasining i-qatorini B matritsasining j-ustuni ko‘paytirish; (2) ustunlar bo‘yicha — $AB = [Ab_1 ... Ab_p]$; (3) qatorlar bo‘yicha — $ABhar$ bir qatori B qatorlarining kombinatsiyasi; (4) tashki ko‘paytmalar bo‘yicha — $AB = \sum_k a_k \cdot r_k$($A$ matritsasining k-ustuni B matritsasining k-qatoriga ko‘paytirish).

$$(AB)C = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC, AI = IA = A$$

$$umumanolgandaAB \neq BA; (A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

Nocommutativlikning aniq misoli: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, lekin $BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ — gorizontal va vertikal siljishlar (shears) qo‘llanish tartibiga bog‘liq.

3.5.3 Maxsus matritsalar va chiqarib tashlash matritsalari

Birlik matritsa I vektorlarni o‘zgarishsiz qoldiradi; diagonal $D = diag(d_1, ..., d_n)$ koordinatalarni masshtablaydi. Yuqori uchburchak matritsalarda diagonal ostida nollar, pastki uchburchaklarda — diagonal ustida; uchburchak tizimlar to‘g‘ri/teskari almashtirish bilan yechiladi. Chiqarib tashlash amaliyoti — o‘zi matritsa: $R_2 \leftarrow R_2 - 3R_1 E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ ga to‘g‘ri keladi, va $E \cdot \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$; o‘zgargan tizim EAx $= Eb$, va $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ asl qatorlarni tiklaydi.

3.5.4 Ters matritsa: ta’rif, mezonlar, qoidalar

$$AA^{-1} = A^{-1}A = I \to Ax = bbarchabuchunyechimgaegax = A^{-1}b$$

Ters matritsa chiziq o‘tkazuvchi aynalishni teskarisiga o‘zgartiradi. Agar $Ax = 0$ nol bo‘lmagan x uchun, u holda teskari matritsa mavjud emas. Kvadrat A uchun quyidagi bayonotlar teng: teskari matritsa mavjud $\Leftrightarrow$ barcha asosiy elementlar noldan farqli (mumkin bo‘lgan almashtirishdan keyin) $\Leftrightarrow$ ustunlar chiziqli mustaqil $\Leftrightarrow$ $Ax = 0$ faqat $x = 0da \Leftrightarrow Ax = b$ har qanday b uchun yagona yechimga ega.

$$(AB)^{-1} = B^{-1}A^{-1}, (A^T)^{-1} = (A^{-1})^T, (A^{-1})^{-1} = A,$$
$$(A^k)^{-1} = (A^{-1})^k$$

Ko‘paytmani teskarisiga olishda ko‘paytuvchilar tartibi o‘zgaradi: $(AB)(B^{-1}A^{-1}) = I$.

3.5.52×2 teskari matritsa va Gauss-Jordan usuli

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} d & b \\ c & a \end{bmatrix} = (ad - bc) \cdot I \to$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (1/(ad - bc)) \cdot \begin{bmatrix} d & b \\ c & a \end{bmatrix} ad - bc \neq 0bo'lganda$$

Misol: $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$, chunki det $= 1$. Gauss-Jordan usuli: $[A|I] \to [I|A^{-1}]$ ikki yarmini ham bir xil qator amaliyotlari bilan: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{bmatrix}$, ya’ni $\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. Ters matritsaning ustunlari: agar $A^{-1} = [x_1 ... x_n], u$ holda $Ax_j = e_j$ — teskari matritsani topish n ta tizimni yechishni anglatadi; bitta tizim $Ax = b$ uchun uchburchak chiqarib tashlash odatda A^{-1} ni aniq hisoblashdan afzalroqdir.

3.5.$6LU$-ajratma va sistemalarni faktorlar orqali yechish

$$A = LU : chiqaruvchilar—birlikpastkiuchburchakLda,$$
$$chiqarishnatijasi—yuqoriuchburchakU$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} (tekshirishko'paytirishbilan)$$

$$E_k \cdots E_2 E_1 A = U \ \to \ A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$$

”Satrning ko‘paytmasini ayirish” operatsiyasining teskari — ”bu ko‘paytmani qayta qo‘shish”; ko‘paytuvchilar tartibi muhim (o‘ng operatsiya birinchi ta’sir ko‘rsatadi).

3×3 misol: $A=\begin{bmatrix}2&1&1\\4&5&4\\-2&2&3\end{bmatrix}$ ko‘paytuvchilar $m_{21}=2,\ m_{31}=-1,\ m_{32}=1$ bilan $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$ $vaU=\begin{bmatrix}2&1&1\\0&3&2\\0&0&2\end{bmatrix}$ beradi ($LU=Ani$ qator bo‘yicha tekshiring). Sistemani yechish: avval $Ly=b$ ni to‘g‘ri almashtirish bilan, keyin $Ux=y$ ni teskari; $L=\begin{bmatrix}1&0\\3&1\end{bmatrix}$, $U=\begin{bmatrix}2&1\\0&2\end{bmatrix}$, $b=[5$;

19] uchun $y=\begin{bmatrix}5\\4\end{bmatrix}$ $vax=[\frac{3}{2}$; 2] hosil bo‘ladi. Ani bir marta ajratib, ko‘p o‘ng tomonli sistemalarni yechish mumkin.

3.5.$7LDU$-ajratma, asosiy elementlar va determinant

$$U=D\cdot(D^{-1}U),Dasosiyelementlarnio'zichigaoladiU;$$
$$A=L\cdot D\cdot(D^{-1}U)birlikdiagonallarbilanLva$$

$$D^{-1}U$$

$$\begin{bmatrix}2&1\\6&5\end{bmatrix}=\begin{bmatrix}1&0\\3&1\end{bmatrix}\cdot\begin{bmatrix}2&0\\0&2\end{bmatrix}\cdot[1\frac{1}{2};$$
$$01]$$

Asosiy elementlar va determinant: $A=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ uchun $a\neq 0$ asosiy elementlar $avad-cb/a$ ga teng, ularning ko‘paytmasi ad $-bc=\det A$ ga teng. Satrlarni almashtirmasdan determinant asosiy elementlarning ko‘paytmasiga teng; normalangan LDU barcha asosiy elementlar noldan farqli bo‘lganda yagona bo‘ladi.

3.5.8 Permutatsiyalar, $PA=LU$ va qisman tanlov

Nima uchun satrlarni almashtirish kerak: $A=\begin{bmatrix}0&2\\3&4\end{bmatrix}$ teskari, lekin birinchi asosiy element nolli; satrlarni almashtirish $PA=\begin{bmatrix}3&4\\0&2\end{bmatrix}$ beradi — nolli asosiy element satrlarning noto‘g‘ri tartibidan, singanlikdan emas, balki signal bo‘lishi mumkin. Permutatsiya matritsalari I satrlarini almashtiradi; har bir qator va ustunda aynan bitta birlik bor. $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$ satrlarni almashtiradi: PA — A satrlari, AP — A ustunlari; n

satr uchun jami $n!$ permutatsiyalar. Misol: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $(x_1, x_2, x_3)^T ni (x_2, x_3, x_1)^T$ ga o‘tkazadi va vektorlarning uzunliklari va skalyar ko‘paytmalarni saqlaydi.

$$P^{-1} = P^T (chunki P^T \cdot P = I); almashtirishlar PA = LU$$

Misol: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $L = I$, $U = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; almashtirishlarda L dagi ko‘paytuvchilarni to‘g‘ri kuzatib borish muhim. Asosiy elementning qisman tanlovi (partial pivoting): $[10^{-8}1;\ 1\ 1]$ singan emas, lekin juda kichik birinchi asosiy element 10^8 ko‘paytuvchini beradi — satrlarni almashtirish budan qochadi; qoida — faqat ustundagi (asosiy element va uning ostidagilar) modul bo‘yicha eng katta elementni olish.

3.5.9 Transponiranіya, simmetrik matritsalar, LDL^T

$$(A^T)_{ij} = a_{ji}, (A+B)^T = A^T + B^T, (AB)^T = B^T \cdot A^T$$

Misol: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. Simmetrik matritsa $A^T = A$ shartni qanoatlantiradi. Agar simmetrik A matritsa almashtirmasiz LMD-ajralishga ega bo‘lsa, uning normalizlangan yuqori ko‘paytchisi L^T ga teng: $A = LDL^T$; masalan $\begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

3.5.10 Qo‘llanilishi: farqlar, yig‘indilar, uchdiagonal tizimlar

Farq matritsasi $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ $Ax = (x_1, x_2 - x_1, x_3 - x_2)^T$ beradi; uning teskari $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ yig‘indilarni to‘plashni beradi — differensial va integrallashning diskret analogi. Uchdiagonal $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ eng yaqin qo‘shnilar aloqalarini (chekli farq tenglamalari) tasvirlaydi; uning omillari tor lentani saqlaydi va LU orqali yechish tuzilmadan foydalanib, zich teskari matritsadan qochadi.

3.5.11 Hafta xulosi

- Matritsa ko‘paytirish bir nechta teng ma’noli talqinlarga ega, ammo

kommutativ emas.

- Teskari matritsa faqoq noaniq kvadrat matritsalar uchun mavjud. • Istisqo $LMD = LMD$ ajralishida kodlangan. • Qatorlarni almashtirish P matritsasi bilan kodlanadi: $PA = LU$. • Transponirlash va almashtirish 7-hafta vektor fazolariga tayyorlik.

[Manba: 6-hafta dars mashqlari (Yechilgan) — 20 masala yechimlari, 2×2 dan ilg‘or ajralishlargacha]

Mashq 1. 2×2 **teskari matritsa.** $A=\begin{bmatrix}2&1\\2&3\end{bmatrix}$ **uchun** A^{-1} **ni toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A=2\cdot3-1\cdot2=4\neq0$. 2×2 formulasi bo‘yicha: $A^{-1}=(\frac14)\cdot\begin{bmatrix}3&-1\\-2&2\end{bmatrix}=[\frac34-\frac14;$ $-1/2$ $1/2]$. Tekshirish: $A\cdot A^{-1}=\begin{bmatrix}2&1\\2&3\end{bmatrix}\cdot[\frac34-\frac14;\ -\frac12\,\frac12]=[\frac32-\frac12,-\frac12+\frac12;\ \frac32-\frac32,-\frac12+\frac32]=I$.

Mashq 2. 2×2 **tizimni yechish.** $Ax=b$ **ni yeching, bunda** $A=\begin{bmatrix}3&2\\1&4\end{bmatrix}$, $b=\begin{bmatrix}8\\6\end{bmatrix}$**.**

Yechim. Tizma: 3x + 2y $=8, x+$ 4y $=6$. Ikkinchi tenglamani 3 ga ko‘paytirib birinchisini ayirsak, 10y $=10$, ya’ni $y=1, x=6-4=2$. Shunday qilib, $x=\begin{bmatrix}2\\1\end{bmatrix}$. Tekshirish: $A\cdot x=[3\cdot2+2;\ 2+4]=\begin{bmatrix}8\\6\end{bmatrix}=b$.

Mashq 3. Teskari 2×2**.** $A=\begin{bmatrix}4&1\\2&5\end{bmatrix}$ **uchun** A^{-1} **ni toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A=4\cdot5-1\cdot2=18\neq0$. $A^{-1}=(\frac{1}{18})\cdot\begin{bmatrix}5&-1\\-2&4\end{bmatrix}=[\frac{5}{18}-\frac{1}{18};\ -1/9\ \ 2/9]$. Tekshirish: $A\cdot A^{-1}=I$ — masalan, (1,1) elementi $4\cdot(\frac{5}{18})-\frac19=\frac{20}{18}-\frac{2}{18}=1$, $(2,2)$ elementi $-\frac{2}{18}+\frac{20}{18}=1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 4. 2×2 **tizmani yechish.** $Ax=b$ **ni yeching, bunda** $A=\begin{bmatrix}2&2\\1&6\end{bmatrix}$, $b=\begin{bmatrix}4\\7\end{bmatrix}$**.**

Yechim. Tizma: 2x + 2y $=4$ (ya’ni $x+y=2$), $x+$ 6y $=7$. Ayirish: 5y $=5, y=1, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}2&+&2\\1&+&6\end{bmatrix}=\begin{bmatrix}4\\7\end{bmatrix}=b$.

Mashq 5. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}2&1\\2&4\end{bmatrix}$ **uchun.**

Yechim. Asosiy element 2, ko‘paytchi $m_{21}=\frac22=1$: $R_2\leftarrow R_2-R_1U=\begin{bmatrix}2&1\\0&3\end{bmatrix}$ beradi. Ko‘paytchni L ga yozamiz: $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}$, $A=LU$ (tekshirish: $\begin{bmatrix}2&1\end{bmatrix}+\begin{bmatrix}0&3\end{bmatrix}=\begin{bmatrix}2&4\end{bmatrix}$). Asosiy elementlar $2va3niD=\begin{bmatrix}2&0\\0&3\end{bmatrix}$ ga chiqaramiz; U ni satrlarni asosiy elementlarga bo‘lish orqali normallasak, $D^{-1}U=[1\frac12;\ 0\ 1]$ hosil bo‘ladi. Shunday qilib, $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 6. 2×2 **ajratma.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}3&1\\3&5\end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{3}{3} = 1$: $R_2 \leftarrow R_2 - R_1U = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$ beradi. $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A = LU$ (tekshirish: $\begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$). $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{3};\ 0\ 1]$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 7. 2×2 **ajratma.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1U = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ beradi. $L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$, $A = LU$ (tekshirish: $4 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \end{bmatrix}$). Ikkinchi asosiy element manfiy — bu ruxsat etilgan: $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $D^{-1}U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 8. 2×2 **ajratma.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1U = [21;\ 0\ 3/2]$ beradi $(4 - \frac{5}{2} = \frac{3}{2})$. $L = [10;\ \frac{5}{2}1], A = LU$ (tekshirish: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + [0\frac{3}{2}] = \begin{bmatrix} 5 & 4 \end{bmatrix}$). $D = [20;\ 0\frac{3}{2}]$, $D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 9. Chiqarish 3×3**.** $Ax = b$ **ni Gauss chiqarish usuli bilan yeching** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0|-2)$; $R_3 - R_1 = (0, 1, 2|0)$. 2-qadam $(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2|-2)$. Uchburchak tizma: $x+$ 3y $+z = 6, -y = -2$, 2z $= -2$. Orqali almashtirish: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b.$

Mashq 10. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$ **uchun** $Ax = b$ **tenglamani Gauss usulida chiqarib tashlash orqali yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 3|-1)$. 2-qadam $(m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3|-3)$. Uchburchak tizim: $x + y + z = 2, y = 2$, 3z $= -3$. Orqali almashtirish: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b.$

Mashq 11. 3×3 **chiqarib tashlash (noaniy holat).** $A=\begin{bmatrix}1&2&1\\2&4&2\\1&3&5\end{bmatrix}$, $b=\begin{bmatrix}4\\8\\2\end{bmatrix}$ **uchun** $Ax=b$ **tenglamani Gauss usulida chiqarib tashlash orqali yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&2&1&4\\2&4&2&8\\1&3&5&2\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,0,0|0)$ — ikkinchi tenglama birinchisining aniq ikki barobariga teng; $R_3-R_1=(0,1,4|-2)$. $2va3$-qatorlarni almashtirgandan keyin: $\begin{bmatrix}1&2&1&4\\0&1&4&-2\\0&0&0&0\end{bmatrix}$ — uchta noma'luf uchta ustunli element, bir o'zgaruvchi erkin. $z=t$ deb olamiz: $y=-2-$ 4t $(y+4z=-2dan), x=4-$ 2y $-z=8+$ 7t. Umumiy yechim $x=\begin{bmatrix}8&+&7t\\-2&&4t\\t&&\end{bmatrix}=\begin{bmatrix}8\\-2\\0\end{bmatrix}+t\cdot\begin{bmatrix}7\\-4\\1\end{bmatrix}, t\in R$ — cheksiz ko'p yechimlar; faylda ko'rsatilgan $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$ — $t=-1$ bo'lganda maxsus yechim. Tekshirish: $A\cdot\begin{bmatrix}1\\2\\-1\end{bmatrix}=\begin{bmatrix}1&+&4&1\\2&+&8&2\\1&+&6&5\end{bmatrix}=\begin{bmatrix}4\\8\\2\end{bmatrix}=b$. Geometrik jihatdan: birinchi ikki tenglama bitta tekislikni belgilaydi, uning uchinchi bilan kesishishi to'g'ri chiziq bo'ladi.

Mashq 12. 3×3 **chiqarib tashlash.** $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&4\end{bmatrix}$, $b=\begin{bmatrix}6\\10\\5\end{bmatrix}$ **uchun** $Ax=b$ **tenglamani Gauss usulida chiqarib tashlash orqali yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&4&5\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,3|-3)$. Uchburchak tizim: $x+$ 3y $+z=6, -y=-2,$ 3z $=-3$. Orqali almashtirish: $z=-1, y=2, x=6-6+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1&+&6&1\\2&+&10&2\\1&+&8&4\end{bmatrix}=\begin{bmatrix}6\\10\\5\end{bmatrix}=b$.

Mashq 13. $LUvaLDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&3&2\\-2&1&1\end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. Ko'paytirgichlar $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$.

Ikkinchi ustun elementi $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. Shunday qilib, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: $LU = A$ (uchinchi qator: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, normalizatori yuqori ko‘paytma $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 14. $LUvaLDU 3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0,3,1)$; $R_3 + R_1 = (0,6,4)$. Ikkinchi ustun elementi $3, m_{32} = 2$: $R_3 - 2R_2 = (0,0,2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; tekshirish: LU ning uchinchi qatori $\begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$ ga teng. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 15. $LUvaLDU 3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0,4,1)$; $R_3 + R_1 = (0,12,6)$. Ikkinchi ustun elementi $4, m_{32} = 3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; tekshirish: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 16. $LUvaLDU 3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Ikkinchi ustun elementi $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: ikkinchi qator $2 \cdot \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 3 \end{bmatrix}$, uchinchisi $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} =$

$\begin{bmatrix}-2 & 1 & 1\end{bmatrix}$. $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 17. Qatorlarni almashtirish va PLU. $A = \begin{bmatrix}0 & 2 & 1\\ 1 & 1 & 0\\ 2 & 3 & 2\end{bmatrix}$ **uchun $PA = LU$ tenglamani qanoatlantiruvchi qatorlarni almashtirish P va ko‘paytuvchilarni toping.**

Yechim. Birinchi diagonal element nol — $1va2$-qatorlarni almashtiramiz: $P = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$, $PA = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix}$. Chiqarish: (2, 1) element allaqachon nol $(m_{21} = 0)$; $m_{31} = 2$: $R_3 - 2R_1 = (2,\,3,\,2) - 2 \cdot (1,\,1,\,0) = (0,\,1,\,2)$; keyin $m_{32} = \frac{1}{2}$: $R_3 - (\frac{1}{2})R_2 = (0,\,1,\,2) - (0,\,1,\,\frac{1}{2}) = (0,\,0,\,\frac{3}{2})$. Shunday qilib, $L = [100$; 0 1 0; $2\,\frac{1}{2}\,1], U = [110$; 0 2 1; 0 0 3/2]. Tekshirish: $LU = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix} = PA$ (uchinchi qator: $2 \cdot (1,\,1,\,0) + (\frac{1}{2}) \cdot (0,\,2,\,1) + (0,\,0,\,\frac{3}{2}) = (2,\,3,\,2)$). Demak, $PA = LU$: P tenglamalarni qayta tartiblab, mos asosiy element beradi; $L2va\frac{1}{2}$ ko‘paytirlarini asl belgilari bilan o‘z ichiga oladi, garchi amallar ularni ayirib tashlasa.

Mashq 18. Transponiranya va teskari. $A = \begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 2 & 0 & 1\end{bmatrix}$ **uchun A^T hisoblang va $(A^T)^{-1} = (A^{-1})^T$ tekshiring.**

Yechim. Transponiranya: $A^T = \begin{bmatrix}1 & 0 & 2\\ 2 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$. $\det A = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5$. Qo‘shimcha matritsa orqali $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix}1 & 2 & -2\\ -2 & 1 & 4\\ 2 & -1 & 1\end{bmatrix}$. A^T dan mustaqil ravishda $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix}1 & -2 & 2\\ 2 & 1 & -1\\ -2 & 4 & 1\end{bmatrix}$ hisoblaymiz; keyin $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix}1 & 2 & -2\\ -2 & 1 & 4\\ 2 & -1 & 1\end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T \cdot (A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix}2 & 2 & 0\\ 2 & 5 & 1\\ 0 & 1 & 3\end{bmatrix}$ $niLDL^T$ ko‘rinishida ajrating.

Yechim. Asosiy element 2; $m_{21} = \frac{2}{2} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Ikkinchi asosiy element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0, 1, 3) - (0, 1, \frac{1}{3}) = (0, 0, \frac{8}{3})$. Almashtirishsiz simmetrik A uchun normalizlangan yuqori ko‘paytirchi L^T ga teng, ya’ni $A = LDL^T$, bunda $L = [100$;

1 1 0; $0\,\frac{1}{3}\,1]$, $D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$. Tekshirish: $L\cdot D = [200;\ 2\ 3\ 0;\ 0\ 1\ 8/3]$, va $(LD)\cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal model. $Ax = b$ni LU **orqali yechishing** $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$**. Eng yaqin qo‘shnilar tuzilishini talqin qiling.**

Yechim. Ajratish: $m_{21} = -\frac{1}{2}$, shuning uchun $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1,\ 2,\ -1) + (1,\ -\frac{1}{2},\ 0) = (0,\ \frac{3}{2},\ -1)$; $m_{32} = -\frac{2}{3}$, shuning uchun $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0,\ -1,\ 2) + (0,\ 1,\ -\frac{2}{3}) = (0,\ 0,\ \frac{4}{3})$. Shunday qilib, $L = [100;\ -1/2\ 1\ 0;\ 0\ -\frac{2}{3}\,1]$, $U = [2\ -1 0;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$. To‘g‘ri almashtirish $Ly = b$: $y_1 = 2$; $-(\frac{1}{2})\cdot 2 + y_2 = 0$, bundan $y_2 = 1$; $-(\frac{2}{3})\cdot 1 + y_3 = 2$, bundan $y_3 = \frac{8}{3}$. Orqaga almashtirish $Ux = y$: $(\frac{4}{3})x_3 = \frac{8}{3}$, $x_3 = 2$; $(\frac{3}{2})x_2 - 2 = 1$, $x_2 = 2$; $2x_1 - 2 = 2, x_1 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 & 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. Matritsa uch diagonal — nol bo‘lmagan elementlar faqat asosiy diagonal va unga qo‘shni pozitsiyalarda: har bir noma’lum faqat eng yaqin qo‘shnilar bilan bog‘langan.

[Manba: 6-hafta Amaliy mashq mashqlari bilan Yechimlar — 20 masala; Tanlangan batafsil yechimlar — masalalar 9–12, 17,

20 ning batafsil versiyalari]

Mashq 1. 2×2 **teskari.** $A = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$ **uchun** A^{-1} **toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A = 3\cdot 6 - 2\cdot 1 = 16 \neq 0$. $A^{-1} = (\frac{1}{16})\cdot \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix} = [\frac{3}{8} - \frac{1}{8};\ -1/16\ 3/16]$. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{9}{8} - \frac{2}{16} = 1, (2,2)$ elementi $-\frac{1}{8} + \frac{18}{16} = 1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 2. 2×2 **tizimni yechish.** $Ax = b$ **ni yeching, bunda** $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 14 \\ 12 \end{bmatrix}$**.**

Yechim. Tizim: $4x + y = 14$, $2x + 3y = 12$. Birinchi tenglamani 3 ga ko‘paytirib, ikkinchisini ayirsak, $10x = 30$, ya’ni $x = 3, y = 14 - 12 = 2$. Shunday qilib, $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Tekshirish: $A\cdot x = \begin{bmatrix} 12 & + & 2 \\ 6 & + & 6 \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \end{bmatrix} = b$.

Mashq 3. 2×2 **teskari.** $A = \begin{bmatrix} 2 & 2 \\ 1 & 4 \end{bmatrix}$ **uchun** A^{-1} **toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A = 2\cdot 4 - 2\cdot 1 = 6 \neq 0$. $A^{-1} = (\frac{1}{6})\cdot \begin{bmatrix} 4 & -2 \\ -1 & 2 \end{bmatrix} = [\frac{2}{3} - \frac{1}{3};\ -1/6\ 1/3]$. Tekshirish: $A\cdot A^{-1} = I$ — (1,1) elementi $\frac{4}{3} - \frac{1}{3} = 1, (2,2)$ elementi $-\frac{1}{3} + \frac{4}{3} = 1$, diagonal bo‘lmagan elementlar — nollar.

Mashq 4. 2×2 **tizimni yechish.** $Ax=b$ **ni yeching, bunda** $A=\begin{bmatrix}3&1\\2&5\end{bmatrix}$, $b=\begin{bmatrix}8\\14\end{bmatrix}$**.**

Yechim. Tizim: 3x $+y=8$, 2x + 5y $=14$. Birinchi tenglamani 5 ga ko‘paytirib, ikkinchisini ayirsak, 13x $=26$, ya’ni $x=2, y=8-6=2$. Shunday qilib, $x=\begin{bmatrix}2\\2\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}6&+&2\\4&+&10\end{bmatrix}=\begin{bmatrix}8\\14\end{bmatrix}=b$.

Mashq 5. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}3&1\\5&5\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=\frac{5}{3}$: $R_2\leftarrow R_2-(\frac{5}{3})R_1U=[31;\ 0\ 10/3]$ beradi $(5-\frac{5}{3}=\frac{10}{3})$. $L=[10;\ \frac{5}{3}1], A=LU$ (tekshirish: $(\frac{5}{3})\cdot\begin{bmatrix}3&1\end{bmatrix}+[0\frac{10}{3}]=\begin{bmatrix}5&5\end{bmatrix}$). $D=[30;\ 0\frac{10}{3}]$, $D^{-1}U=[1\frac{1}{3};\ 0\ 1]$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 6. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}1&1\\2&3\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=2$: $R_2\leftarrow R_2-2R_1danU=\begin{bmatrix}1&1\\0&1\end{bmatrix}$ hosil bo‘ladi. $L=\begin{bmatrix}1&0\\2&1\end{bmatrix}$, $A=LU$ (tekshirish: $2\cdot\begin{bmatrix}1&1\end{bmatrix}+\begin{bmatrix}0&1\end{bmatrix}=\begin{bmatrix}2&3\end{bmatrix}$). Ikkala asosiy element ham 1 ga teng, shuning uchun $D=IvaD^{-1}U=U$: $A=\begin{bmatrix}1&0\\2&1\end{bmatrix}\cdot\begin{bmatrix}1&0\\0&1\end{bmatrix}\cdot\begin{bmatrix}1&1\\0&1\end{bmatrix}$.

Mashq 7. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}2&1\\3&4\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=\frac{3}{2}$: $R_2\leftarrow R_2-(\frac{3}{2})R_1danU=[21;\ 0\frac{5}{2}](4-\frac{3}{2}=\frac{5}{2})$ hosil bo‘ladi. $L=[10;\ \frac{3}{2}1], A=LU$ (tekshirish: $(\frac{3}{2})\cdot\begin{bmatrix}2&1\end{bmatrix}+[0\frac{5}{2}]=\begin{bmatrix}3&4\end{bmatrix}$). $D=[20;\ 0\frac{5}{2}]$, $D^{-1}U=[1\frac{1}{2};\ 0\ 1]$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 8. 2×2 **ajratish.** $A=LU=LDU$ **ni toping** $A=\begin{bmatrix}3&1\\4&5\end{bmatrix}$ **uchun.**

Yechim. $m_{21}=\frac{4}{3}$: $R_2\leftarrow R_2-(\frac{4}{3})R_1danU=[31;\ 0\frac{11}{3}](5-\frac{4}{3}=\frac{11}{3})$ hosil bo‘ladi. $L=[10;\ \frac{4}{3}1], A=LU$ (tekshirish: $(\frac{4}{3})\cdot\begin{bmatrix}3&1\end{bmatrix}+[0\frac{11}{3}]=\begin{bmatrix}4&5\end{bmatrix}$). $D=[30;\ 0\frac{11}{3}]$, $D^{-1}U=[1\frac{1}{3};\ 0\ 1]$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 9. 3×3 **chiqarib tashlash.** $Ax=b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A=\begin{bmatrix}1&1&1\\2&3&2\\1&2&4\end{bmatrix}$, $b=\begin{bmatrix}2\\6\\1\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+y+z=2$, 2x + 3y + 2z $=6, x+$ 2y + 4z $=1$; kengaytirilgan matritsa $\begin{bmatrix}1&1&1&2\\2&3&2&6\\1&2&4&1\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21}=2, m_{31}=1)$: yangi ikkinchi qator (2, 3, $2|6)-2\cdot(1,\ 1,\ 1|2)=(0,\ 1,\ 0|2)$; yangi uchinchi qator $(1,\ 2,\ 4|1)-(1,\ 1,\ 1|2)=(0,\ 1,\ 3|-1)$. y ni chiqarib tashlaymiz $(m_{32}=1)$: $(0,\ 1,\ 3|-1)-(0,\ 1,\ 0|2)=(0,\ 0,\ 3|-3)$. Uchburchak

tizma: $x+y+z=2, y=2$, 3z $=-3$. Orqali almashtirish: $z=-1, y=2, x=2-2+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1&+&2&1\\2&+&6&2\\1&+&4&4\end{bmatrix}=\begin{bmatrix}2\\6\\1\end{bmatrix}=b$.

Mashq 10. 3×3 **chiqarib tashlash (cheksiz ko‘p yechimlar).** $Ax=b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A=\begin{bmatrix}1&2&1\\2&4&2\\1&3&5\end{bmatrix}$, $b=\begin{bmatrix}4\\8\\2\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+$ 2y $+z=4$, 2x + 4y + 2z $=8, x+$ 3y + 5z $=2$; kengaytirilgan matritsa $\begin{bmatrix}1&2&1&4\\2&4&2&8\\1&3&5&2\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21}=2, m_{31}=1)$: ikkinchi qator (0, 0, 0 | 0) beradi — $0=0$ tenglama cheklovlar qoymaydi (ikkinchi tenglama birinchisining aniq ikki barobaridir); uchinchi qator: (0, 1, 4 | −2). $2va3$-qatorlarni almashtirib, bosqichli ko‘rinishga $\begin{bmatrix}1&2&1&4\\0&1&4&-2\\0&0&0&0\end{bmatrix}$ kelib chiqadi — uchta noma’lum uchun ikkita asosiy element. $z=t, t\in R$ deb olaylik: ikkinchi qatordan $y=-2-$ 4t; birinchidan $x=4-$ 2y $-z=8+$ 7t. Barcha yechimlar: $x=\begin{bmatrix}8&+&7t\\-2&&4t\\t&&\end{bmatrix}=\begin{bmatrix}8\\-2\\0\end{bmatrix}+t\cdot\begin{bmatrix}7\\-4\\1\end{bmatrix}$, $t\in R$. Tekshirish: $(8+7t)+2(-2-4t)+t=4va(8+7t)+3(-2-4t)+$ 5t $=2$; ikkinchi tenglama avtomatik ravishda birinchining ikki barobari sifatida bajariladi. Geometrik jihatdan: birinchi ikki tenglama bir xil tekislikni tasvirlaydi va uning uchinchi bilan kesishishi — to‘g‘ri chiziq (yechimlar oilasi). Qisqa javob fayli $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$ — $t=-1$ da xususiy yechim:

$$A\cdot\begin{bmatrix}1\\2\\-1\end{bmatrix}=\begin{bmatrix}4\\8\\2\end{bmatrix}=b.$$

Mashq 11. 3×3 **chiqarib tashlash.** $Ax=b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&4\end{bmatrix}$, $b=\begin{bmatrix}6\\10\\5\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+$ 3y $+z=6$, 2x + 5y + 2z $=10, x+$ 4y + 4z $=5$; kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&4&5\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21}=2, m_{31}=1)$: ikkinchi qator (0, −1, 0 | −2); uchinchi qator (0, 1, 3 | −1). Ikkinchi asosiy element −1; uning ostidagi elementni yo‘qotish uchun 2-qatorni 3-qatorga qo‘shamiz $(m_{32}=-1)$: $(0,1,3|-1)+(0,-1,0|-2)=(0,0,3|-3)$. Uchburchak tizma: $x+$ 3y $+z=6, -y=-2$, 3z $=-3$. Orqali

almashtirish: $z=-1, y=2, x=6-6+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish:

$$A\cdot x=\begin{bmatrix}1&+&6&1\\2&+&10&2\\1&+&8&4\end{bmatrix}=\begin{bmatrix}6\\10\\5\end{bmatrix}=b.$$

Mashq 12. 3×3 **chiqarib tashlash.** $Ax=b$ **tenglamani Gauss chiqarib tashlash usuli bilan yeching** $A=\begin{bmatrix}1&1&1\\2&3&2\\1&2&3\end{bmatrix}$, $b=\begin{bmatrix}2\\6\\2\end{bmatrix}$ **uchun.**

Yechim. Tizma: $x+y+z=2$, 2x + 3y + 2z = $6, x+$ 2y + 3z = 2; kengaytirilgan matritsa $\begin{bmatrix}1&1&1&2\\2&3&2&6\\1&2&3&2\end{bmatrix}$. x ni chiqarib tashlaymiz $(m_{21}=2, m_{31}=1)$: ikkinchi qator (0, 1, 0 | 2); uchinchi qator (0, 1, 2 | 0). y ni chiqarib tashlaymiz $(m_{32}=1)$: $(0,\ 1,\ 2|0)-(0,\ 1,\ 0|2)=(0,\ 0,\ 2|-2)$. Uchburchak tizma: $x+y+z=2, y=2$, 2z $=-2$. Orqali almashtirish: $z=-1, y=2, x=2-2+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish:

$$A\cdot x=\begin{bmatrix}1&+&2&1\\2&+&6&2\\1&+&4&3\end{bmatrix}=\begin{bmatrix}2\\6\\2\end{bmatrix}=b.$$

Mashq 13. $LUvaLDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&5&3\\-2&5&3\end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi asosiy element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: uchinchi qator

$$LU-\begin{bmatrix}2&1&1\end{bmatrix}+2\cdot\begin{bmatrix}0&3&1\end{bmatrix}+\begin{bmatrix}0&0&2\end{bmatrix}=\begin{bmatrix}-2&5&3\end{bmatrix}.$$ $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=[1\ \frac{1}{2}\ \frac{1}{2}$; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 14. $LUvaLDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&5&2\\-2&11&5\end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Ikkinchi asosiy element $4, m_{32}=3$: $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&3&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&4&1\\0&0&3\end{bmatrix}$; tekshirish:

$-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+3\cdot\begin{bmatrix}0 & 4 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 3\end{bmatrix}=\begin{bmatrix}-2 & 11 & 5\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&4&0\\0&0&3\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 15. $LUvaLDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&4&3\\-2&1&1\end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Ikkinchi asosiy element $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $2\cdot\begin{bmatrix}2 & 1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2 & 1\end{bmatrix}=\begin{bmatrix}4 & 4 & 3\end{bmatrix}$ $va-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+\begin{bmatrix}0 & 2 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 1\end{bmatrix}=\begin{bmatrix}-2 & 1 & 1\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 16. $LUvaLDU3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&4&2\\-2&5&3\end{bmatrix}$ **uchun** $LUvaLDU$ **ni toping.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi asosiy element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: uchinchi qator $LU-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+2\cdot\begin{bmatrix}0 & 3 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 2\end{bmatrix}=\begin{bmatrix}-2 & 5 & 3\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 17. Qatorlarning almashtirishi va PLU**.** $A=\begin{bmatrix}0&2&1\\1&1&0\\2&3&2\end{bmatrix}$ **uchun** $PA=LU$ **tenglikni qanoatlantiruvchi qatorlarning almashtirishi** P **va omillarni toping.**

Yechim. (1) Birinchi diagonal element 0 — uni asosiy element sifatida ishlatib bo‘lmaydi; $1va2$-qatorlarni almashtiramiz. Mos almashtiruvchi matritsa (I da bir xil almashtirish): $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$; A ni chapdan P ga ko‘paytirish qatorlarni almashtiradi: $PA=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}$. (2) Birinchi asosiy element 1 dan pastni chiqaramiz: (2, 1) elementi allaqachon nol ($m_{21}=0$); $m_{31}=\frac{2}{1}=2$, va $R_3\leftarrow R_3-2R_1$: $(2,\,3,\,2)-2\cdot(1,\,1,\,0)=(0,\,1,\,2)$. (3) Ikkinchi asosiy element 2; $m_{32}=\frac{1}{2}$, va $R_3\leftarrow R_3-(\frac{1}{2})R_2$: $(0,\,1,\,2)-(0,\,1,\,\frac{1}{2})=(0,\,0,\,\frac{3}{2})$;

$U = [110;\ 0\ 2\ 1;\ 0\ 0\ 3/2]$. (4) L ni omillardan quramiz: $L = [100;\ m_{21}10;\ m_{31}m_{32}1] = [100;\ 0\ 1\ 0;\ 2\ 1/2\ 1]$. (5) Tekshirish: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$ — uchinchi qator $2\cdot(1,1,0)+(\frac{1}{2})\cdot(0,2,1)+(0,0,\frac{3}{2}) = (2,3,2)$. Shunday qilib, $PA = LU$: P tenglamalarni qayta tartibga solib, mos asosiy elementni ta’minlaydi, chiqarish esa faktorizatsiyani yakunlaydi; $L2va\frac{1}{2}$ omillarini asl belgilari bilan o‘z ichiga oladi, garchi amaliyotlar ularni ayiradi.

Mashq 18. Transponir va teskari. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **uchun** A^T **ni hisoblang va** $(A^T)^{-1} = (A^{-1})^T$ **tekshiring.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. $\det A = 1\cdot(1\cdot1-1\cdot0)-2\cdot(0\cdot1-1\cdot2)+0\cdot(0\cdot0-1\cdot2) = 5$. Qo‘shni matritsa bo‘yicha $(A^T)^{-1} = (\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Alohida: $A^{-1} = (\frac{1}{5})\cdot\begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, bundan $(A^{-1})^T = (\frac{1}{5})\cdot\begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T\cdot(A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T**. Simmetrik** $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ $niLDL^T$ **ko‘rinishida ajrating.**

Yechim. Asosiy element 2; $m_{21} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0,3,1)$. Ikkinchi asosiy element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0,\ 0,\ \frac{8}{3})$. Almashtirishsiz simmetrik A uchun $A = LDL^T$, bu yerda $L = [100;\ 1\ 1\ 0;\ 0\frac{1}{3}1], D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$. Tekshirish: $L\cdot D = [200;\ 2\ 3\ 0;\ 0\ 1\ 8/3]$, va $(LD)\cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal model. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ **uchun** $Ax = bniLU$ **orqali yeching. Eng yaqin qo‘shnilarning tuzilishini tushuntiring.**

Yechim. (1) Ajratish: $m_{21} = (-1)/2 = -\frac{1}{2}$, shuning uchun $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1,\ 2,\ -1) + (1,\ -\frac{1}{2},\ 0) = (0,\ \frac{3}{2},\ -1)$; $m_{32} = (-1)/(\frac{3}{2}) = -\frac{2}{3}$, shuning uchun $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0,\ -1,\ 2) + (0,\ 1,\ -\frac{2}{3}) = (0,\ 0,\ \frac{4}{3})$. Shunday qilib, $L = [100;\ -1/2\ 1\ 0;\ 0 - \frac{2}{3}1], U = [2 - 10;\ 0\ 3/2$

−1; 0 0 4/3]; ko‘paytchilar L da boshlang‘ch ishoralar bilan, minuslarni ham hisobga olgan holda paydo bo‘ladi. (2) To‘g‘ri almashtirish $Ly = b$: $y_1 = 3$; $-(\frac{1}{2}) \cdot 3 + y_2 = 0$, $y_2 = \frac{3}{2}$; $-(\frac{2}{3}) \cdot (\frac{3}{2}) + y_3 = 2$, $y_3 = 3$. (3) Tegishli almashtirish $Ux = y$: $(\frac{4}{3})x_3 = 3$, $x_3 = \frac{9}{4}$; $(\frac{3}{2})x_2 - \frac{9}{4} = \frac{3}{2}$, $x_2 = (\frac{15}{4}) \cdot (\frac{2}{3}) = \frac{5}{2}$; $2x_1 - \frac{5}{2} = 3$, $x_1 = \frac{11}{4}$. Shunday qilib, $x = [\frac{11}{4}$; 5/2; 9/4]. Tekshirish: $2 \cdot (\frac{11}{4}) - \frac{5}{2} = 3$; $-(\frac{11}{4}) + 2 \cdot (\frac{5}{2}) - \frac{9}{4} = 0$; $-(\frac{5}{2}) + 2 \cdot (\frac{9}{4}) = 2$. (4) Eng yaqin qo‘shnilar tuzilishi: matritsa uchdiagonaldir (nol bo‘lmagan elementlar faqat asosiy diagonalda va uning ustida va ostida joylashgan); $x_2 x_1 va x_3$ bilan o‘zaro ta’sir qiladi, $x_1 - x_3$ o‘rtasida to‘g‘ridan-to‘g‘ri bog‘liqlik yo‘q $(a_{13} = a_{31} = 0)$. Bunday tuzilma sterjen bo‘ylab issiqlik o‘tkazuvchanlik modellarida, prujinalar zanjirida va bir o‘lchamli chekli farqli tenglamalarda paydo bo‘ladi (A — qadam oralig‘i multiplikatori gacha bo‘lgan standart diskret manfiy ikkinchi hosila); chiqarib tashlash tor tuzilimni saqlaydi (L va U ikkidiagonal), shuning uchun sistema arzon hal qilinadi. $(5) LDU$: $D = [200$; 0 3/2 0; $00\frac{4}{3}] va D^{-1}U = [1 - \frac{1}{2}0$; 0 1 −2/3; 0 0 1], shuning uchun $A = L \cdot D \cdot D^{-1}U$.

[Manba: 6-hafta Mustaqil O‘qitish Topshirig‘i — 40 ta taklif qilingan masala (asosiy 2×2 dan tortib, qiyin masalagacha);

ko‘rsatmalar: chiqarish bosqichlarini ko‘rsatish, ko‘paytmalarni tekshirish va teskari matritsa yoki ajratish mavjudligini asoslash.

Yechimlar Python da tekshirilgan]

Mashq 1. 2×2 **teskari matritsa.** $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$ **uchun** A^{-1} **ni toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A = 2 \cdot 6 - 2 \cdot 1 = 10 \neq 0$. $A^{-1} = (\frac{1}{10}) \cdot \begin{bmatrix} 6 & -2 \\ -1 & 2 \end{bmatrix} = [\frac{3}{5} - \frac{1}{5}$; −1/10 1/5]. Tekshirish: $A \cdot A^{-1} = I$ — (1,1) elementi $\frac{6}{5} - \frac{1}{5} = 1, (2,2)$ elementi $-\frac{1}{5} + \frac{6}{5} = 1$, diagonal bo‘lmagan elementlar nol.

Mashq 2. 2×2 **sistemani yechish.** $Ax = b$ **sistemani yeching, bunda** $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$.

Yechim. Sistema: 3x $+y = 8$, 2x + 3y $= 10$. Birinchi tenglamani 3 ga ko‘paytirib, ikkinchisini ayirsak, 7x $= 14$, ya’ni $x = 2, y = (8 - 6)/1 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 6 & + & 2 \\ 4 & + & 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix} = b$.

Mashq 3. 2×2 **teskari matritsa.** $A = \begin{bmatrix} 4 & 2 \\ 1 & 4 \end{bmatrix}$ **uchun** A^{-1} **ni toping. Ko‘paytirish orqali tekshiring.**

Yechim. $\det A = 4 \cdot 4 - 2 \cdot 1 = 14 \neq 0$. $A^{-1} = (\frac{1}{14}) \cdot \begin{bmatrix} 4 & -2 \\ -1 & 4 \end{bmatrix} = [\frac{2}{7} - \frac{1}{7}$; −1/14 2/7]. Tekshirish: $A \cdot A^{-1} = I$ — (1,1) elementi $\frac{8}{7} - \frac{1}{7} = 1, (2,2)$ elementi $-\frac{1}{7} + \frac{8}{7} = 1$, diagonal bo‘lmagan elementlar nol.

Mashq 4. 2×2 **sistemani yechish.** $Ax = b$ **sistemani yeching, bunda** $A = \begin{bmatrix} 2 & 1 \\ 2 & 5 \end{bmatrix}$,

$b = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.

Yechim. Sistema: 2x $+y = 4$, 2x + 5y $= 12$. Ikkinchi tenglamadan birinchisini ayirsak: 4y $= 8, y = 2$; keyin 2x $= 4 - 2, x = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 2 & + & 2 \\ 2 & + & 10 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \end{bmatrix} = b$.

Mashq 5. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1 U = [21;\ 0\ 3/2]$ beradi. $L = [10;\ \frac{5}{2}1], A = LU$ (tekshirish: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + [0\frac{3}{2}] = \begin{bmatrix} 5 & 4 \end{bmatrix}$). $D = [20;\ 0\frac{3}{2}]$, $D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 6. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = \frac{2}{3}$: $R_2 \leftarrow R_2 - (\frac{2}{3})R_1 U = [31;\ 0\ 13/3]$ beradi $(5 - \frac{2}{3} = \frac{13}{3})$. $L = [10;\ \frac{2}{3}1], A = LU$ (tekshirish: $(\frac{2}{3}) \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} + [0\frac{13}{3}] = \begin{bmatrix} 2 & 5 \end{bmatrix}$). $D = [30;\ 0\frac{13}{3}]$, $D^{-1}U = [1\frac{1}{3};\ 0\ 1]$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 7. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = 3$: $R_2 \leftarrow R_2 - 3R_1 U = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ beradi. $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $A = LU$ (tekshirish: $3 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 5 \end{bmatrix}$). $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $D^{-1}U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 8. 2×2 **ajratish.** $A = LU = LDU$ **ni toping** $A = \begin{bmatrix} 2 & 1 \\ 4 & 4 \end{bmatrix}$ **uchun.**

Yechim. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1 U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ beradi. $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $A = LU$ (tekshirish: $2 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $D^{-1}U = [1\frac{1}{2};\ 0\ 1]$, va $A = L \cdot D \cdot D^{-1}U$.

Mashq 9. 3×3 **chiqarish.** $Ax = b$ **ni Gauss chiqarish usuli bilan yeching** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0|-2)$; $R_3 - R_1 = (0, 1, 3|-1)$. 2-qadam $(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3|-3)$. Uchburchak tizim: $x+$ 3y $+z = 6, -y = -2$, 3z $= -3$; bundan $z = -1, y = 2, x = 1$. Shunday qilib, $x =$

$\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1&+&6&1\\2&+&10&2\\1&+&8&4\end{bmatrix}=\begin{bmatrix}6\\10\\5\end{bmatrix}=b$.

Mashq 10. 3×3 **chiqarish.** $Ax=b$ **ni Gauss chiqarish usuli bilan yeching** $A=\begin{bmatrix}1&1&1\\2&3&2\\1&2&3\end{bmatrix}$, $b=\begin{bmatrix}2\\6\\2\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&1&1&2\\2&3&2&6\\1&2&3&2\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,1,0|2)$; $R_3-R_1=(0,1,2|0)$. 2-qadam $(m_{32}=1)$: $R_3-R_2=(0,0,2|-2)$. Uchburchak tizim: $x+y+z=2, y=2$, 2z $=-2$; bundan $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$.

Tekshirish: $A\cdot x=\begin{bmatrix}1&+&2&1\\2&+&6&2\\1&+&4&3\end{bmatrix}=\begin{bmatrix}2\\6\\2\end{bmatrix}=b$.

Mashq 11. 3×3 **chiqarib tashlash (cheksiz ko‘p yechimlar).** $Ax=b$ **tenglamani Gaus usulida chiqarib tashlash orqali yeching** $A=\begin{bmatrix}1&2&1\\2&4&2\\1&3&6\end{bmatrix}$, $b=\begin{bmatrix}4\\8\\1\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&2&1&4\\2&4&2&8\\1&3&6&1\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,0,0|0)$ — ikkinchi tenglama birinchisining ikki barobar katta va cheklovlar qo‘shmaydi; $R_3-R_1=(0,1,5|-3)$. $2va3$-qatorlarni almashtirgandan keyin: $\begin{bmatrix}1&2&1&4\\0&1&5&-3\\0&0&0&0\end{bmatrix}$ — uchta noma’lum uchun ikkita asosiy element. Ozod o‘zgaruvchi $z=t$: $y=-3-$ 5t, $x=4-$ 2y $-z=10+$ 9t. Cheksiz ko‘p yechimlar: $x=\begin{bmatrix}10&+&9t\\-3&&5t\\t&&\end{bmatrix}$, $t\in R$; maxsus yechim $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$ $t=-1$ ga to‘g‘ri keladi. Tekshirish: $A\cdot\begin{bmatrix}1\\2\\-1\end{bmatrix}=\begin{bmatrix}1&+&4&1\\2&+&8&2\\1&+&6&6\end{bmatrix}=\begin{bmatrix}4\\8\\1\end{bmatrix}=b$.

Mashq 12. 3×3 **chiqarib tashlash.** $Ax=b$ **tenglamani Gaus usulida chiqarib**

tashlash orqali yeching $A=\begin{bmatrix}1&3&1\\2&5&2\\1&4&3\end{bmatrix}$, $b=\begin{bmatrix}6\\10\\6\end{bmatrix}$ **uchun.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1&3&1&6\\2&5&2&10\\1&4&3&6\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,2|0)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,2|-2)$. Orqali almashtirish: $z=-1, y=2, x=6-6+1=1$. Shunday qilib, $x=\begin{bmatrix}1\\2\\-1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}1&+&6&1\\2&+&10&2\\1&+&8&3\end{bmatrix}=\begin{bmatrix}6\\10\\6\end{bmatrix}=b$.

Mashq 13. $LUvaLDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&4&3\\-2&1&1\end{bmatrix}$ **uchun** $LUvaLDU$ **hisoblang.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Ikkinchi asosiy element $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $LU=A$ (uchinchi qator: $-\begin{bmatrix}2&1&1\end{bmatrix}+\begin{bmatrix}0&2&1\end{bmatrix}+\begin{bmatrix}0&0&1\end{bmatrix}=\begin{bmatrix}-2&1&1\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 14. $LUvaLDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\2&4&2\\-2&5&3\end{bmatrix}$ **uchun** $LUvaLDU$ **hisoblang.**

Yechim. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,3,1)$; $R_3+R_1=(0,6,4)$. Ikkinchi asosiy element $3, m_{32}=2$: $R_3-2R_2=(0,0,2)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&2&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&3&1\\0&0&2\end{bmatrix}$; tekshirish: LU uchinchi qatori $-\begin{bmatrix}2&1&1\end{bmatrix}+2\cdot\begin{bmatrix}0&3&1\end{bmatrix}+\begin{bmatrix}0&0&2\end{bmatrix}=\begin{bmatrix}-2&5&3\end{bmatrix}$. $D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&2\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 15. $LUvaLDU 3\times 3$**.** $A=\begin{bmatrix}2&1&1\\4&6&3\\-2&11&5\end{bmatrix}$ **uchun** $LUvaLDU$ **hisoblang.**

Yechim. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Ikkinchi asosiy

element $4, m_{32} = 3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; tekshirish: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 16. $LUvaLDU3 \times 3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** $LUvaLDU$ **hisoblang.**

Yechim. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Ikkinchi asosiy element $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: $LU = A$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 17. Qatorlarni almashtirish va PLU**.** $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$ **uchun qatorlarni almashtirish** $PvaPA = LU$ **tenglikni qanoatlantiruvchi omillarni toping.**

Yechim. Nol birinchi asosiy element — $1va2$-qatorlarni almashtiring: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. Chiqarib tashlash: $m_{31} = 2(R_3 - 2R_1 = (0,\,1,\,2))$; $m_{32} = \frac{1}{2}(R_3 - (\frac{1}{2})R_2 = (0,\,0,\,\frac{3}{2}))$. $L = [100$; 0 1 0; $2\frac{1}{2}1], U = [110$; 0 2 1; 0 0 3/2]; tekshirish: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$. Shunday qilib, $PA = LU$.

Mashq 18. Transponirovka va teskari. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ **uchun** A^T **ni hisoblang va** $(A^T)^{-1} = (A^{-1})^T$ **tenglikni tekshiring.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $\det A = 1 \cdot (1-0) - 2 \cdot (0-2) + 0 = 5$. $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$.

Alohida: $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, bundan $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T \cdot (A^T)^{-1} = I$.

Mashq 19. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ ni LDL^T ko‘rinishida ajrating.

Yechim. Ko‘paytuvchilar $m_{21} = 1$, $m_{31} = 0$, $m_{32} = \frac{1}{3}$. $L = [100;\ 1\ 1\ 0;\ 0\ \frac{1}{3}\ 1]$, $D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$; simmetrik matritsa uchun almashtirishsiz $A = LDL^T$. Tekshirish: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 20. Masala-muhokama: uch diagonal modeli. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ uchun $Ax = b$ ni LU bilan yeching. Eng yaqin qo‘shnilar tuzilishini tushuntiring.

Yechim. LU: $m_{21} = -\frac{1}{2}$ va $m_{32} = -\frac{2}{3}$, shuning uchun $L = [100;\ -1/2\ 1\ 0;\ 0\ -\frac{2}{3}\ 1]$, $U = [2\ -1\ 0;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$. To‘g‘ri almashtirish $Ly = b$: $y_1 = 2$, $y_2 = 1$, $y_3 = \frac{8}{3}$. Orqa almashtirish $Ux = y$: $x_3 = 2, x_2 = 2, x_1 = 2$. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 & 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. Matritsa uch diagonal: bog‘lanishlar faqat eng yaqin qo‘shnilar bilan (x_2} — x_1} va x_3} bilan; $x_1 - x_3$ o‘rtasida to‘g‘ri bog‘lanish yo‘q, $a_{13} = a_{31} = 0$); $LvaU$ ikki diagonal — tor tuzilish saqlanib qolgan.

Mashq 21. Teskari 2×2. $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ uchun A^{-1} ni toping. Ko‘paytirish orqali tekshiring.

Yechim. $\det A = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix} = [\frac{3}{4}\ -\frac{1}{4};\ -1/2\ 1/2]$. Tekshirish: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \cdot [\frac{3}{4}\ -\frac{1}{4};\ -\frac{1}{2}\ \frac{1}{2}] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

Mashq 22. 2×2 tizimini yechish. $Ax = b$ ni yeching, bunda $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

Yechim. Tizim: $3x + 2y = 8, x + 4y = 6$. Ikkinchi tenglamadan $x = 6 - 4y$; almashtirish:

$3(6-4y)+$ 2y $=8$, ya'ni $18-$ 10y $=8, y=1, x=2$. Shunday qilib, $x=\begin{bmatrix}2\\1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}6&+&2\\2&+&4\end{bmatrix}=\begin{bmatrix}8\\6\end{bmatrix}=b$.

Mashq 23. Teskari 2×2**.** $A=\begin{bmatrix}4&1\\2&5\end{bmatrix}$ **uchun** A^{-1} **ni toping. Ko'paytirish orqali tekshiring.**

Yechim. $\det A=4\cdot 5-1\cdot 2=18\neq 0$. $A^{-1}=(\frac{1}{18})\cdot\begin{bmatrix}5&-1\\-2&4\end{bmatrix}=[\frac{5}{18}-\frac{1}{18}$; −1/9 2/9]. Tekshirish: $A\cdot A^{-1}=I$ — (1,1) elementi $\frac{20}{18}-\frac{2}{18}=1, (2,2)$ elementi $-\frac{2}{18}+\frac{20}{18}=1$, diagonal bo'lmagan elementlar — nol.

Mashq 24. 2×2 **tizimini yechish.** $Ax=b$ **ni yeching, bunda** $A=\begin{bmatrix}2&2\\1&6\end{bmatrix}$, $b=\begin{bmatrix}4\\7\end{bmatrix}$**.**

Yechim. Tizim: 2x + 2y $=4$ (ya'ni $x+y=2), x+$ 6y $=7$. Ayirish: 5y $=5, y=1, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}2&+&2\\1&+&6\end{bmatrix}=\begin{bmatrix}4\\7\end{bmatrix}=b$.

Mashq 25. 2×2 **ajratish.** $A=LU=LDUniA=\begin{bmatrix}2&1\\2&4\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=1$: $R_2\leftarrow R_2-R_1U=\begin{bmatrix}2&1\\0&3\end{bmatrix}$ beradi. $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}$, $A=LU$ (tekshirish: $\begin{bmatrix}2&1\end{bmatrix}+\begin{bmatrix}0&3\end{bmatrix}=\begin{bmatrix}2&4\end{bmatrix}$). $D=\begin{bmatrix}2&0\\0&3\end{bmatrix}$, $D^{-1}U=[1\frac{1}{2}$; 0 1], va $A=L\cdot D\cdot D^{-1}U$.

Mashq 26. 2×2 **ajratish.** $A=LU=LDUniA=\begin{bmatrix}3&1\\3&5\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=1$: $R_2\leftarrow R_2-R_1U=\begin{bmatrix}3&1\\0&4\end{bmatrix}$ beradi. $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}$, $A=LU$ (tekshirish: $\begin{bmatrix}3&1\end{bmatrix}+\begin{bmatrix}0&4\end{bmatrix}=\begin{bmatrix}3&5\end{bmatrix}$). $D=\begin{bmatrix}3&0\\0&4\end{bmatrix}$, $D^{-1}U=[1\frac{1}{3}$; 0 1], va $A=L\cdot D\cdot D^{-1}U$.

Mashq 27. 2×2 **ajratish.** $A=LU=LDUniA=\begin{bmatrix}1&1\\4&3\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=4$: $R_2\leftarrow R_2-4R_1U=\begin{bmatrix}1&1\\0&-1\end{bmatrix}$ beradi. $L=\begin{bmatrix}1&0\\4&1\end{bmatrix}$, $A=LU$ (tekshirish: $4\cdot\begin{bmatrix}1&1\end{bmatrix}+\begin{bmatrix}0&-1\end{bmatrix}=\begin{bmatrix}4&3\end{bmatrix}$). $D=\begin{bmatrix}1&0\\0&-1\end{bmatrix}$ (manfiy asos elementi ruxsat etilgan), $D^{-1}U=\begin{bmatrix}1&1\\0&1\end{bmatrix}$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 28. 2×2 **ajratish.** $A=LU=LDUniA=\begin{bmatrix}2&1\\5&4\end{bmatrix}$ **uchun toping.**

Yechim. $m_{21}=\frac{5}{2}$: $R_2 \leftarrow R_2-(\frac{5}{2})R_1 U=[21;\ 0\ 3/2]$ beradi. $L=[10;\ \frac{5}{2}1], A=LU$ (tekshirish: $(\frac{5}{2})\cdot\begin{bmatrix}2 & 1\end{bmatrix}+[0\frac{3}{2}]=\begin{bmatrix}5 & 4\end{bmatrix}$). $D=[20;\ 0\frac{3}{2}]$, $D^{-1}U=[1\frac{1}{2};\ 0\ 1]$, va $A=L\cdot D\cdot D^{-1}U$.

Mashq 29. 3×3 **chiqarib tashlash.** $A=\begin{bmatrix}1 & 3 & 1\\ 2 & 5 & 2\\ 1 & 4 & 3\end{bmatrix}$, $b=\begin{bmatrix}6\\ 10\\ 6\end{bmatrix}$ **uchun** $Ax=b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 3 & 1 & 6\\ 2 & 5 & 2 & 10\\ 1 & 4 & 3 & 6\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,-1,0|-2)$; $R_3-R_1=(0,1,2|0)$. 2-qadam $(m_{32}=-1)$: $R_3+R_2=(0,0,2|-2)$. Orq almashtirish: $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}6\\ 10\\ 6\end{bmatrix}=b$.

Mashq 30. 3×3 **chiqarib tashlash.** $A=\begin{bmatrix}1 & 1 & 1\\ 2 & 3 & 2\\ 1 & 2 & 4\end{bmatrix}$, $b=\begin{bmatrix}2\\ 6\\ 1\end{bmatrix}$ **uchun** $Ax=b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 1 & 1 & 2\\ 2 & 3 & 2 & 6\\ 1 & 2 & 4 & 1\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,1,0|2)$; $R_3-R_1=(0,1,3|-1)$. 2-qadam $(m_{32}=1)$: $R_3-R_2=(0,0,3|-3)$. Orq almashtirish: $z=-1, y=2, x=1$. Shunday qilib, $x=\begin{bmatrix}1\\ 2\\ -1\end{bmatrix}$. Tekshirish: $A\cdot x=\begin{bmatrix}2\\ 6\\ 1\end{bmatrix}=b$.

Mashq 31. 3×3 **chiqarib tashlash (cheksiz ko‘p yechimlar).** $A=\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 3 & 5\end{bmatrix}$, $b=\begin{bmatrix}4\\ 8\\ 2\end{bmatrix}$ **uchun** $Ax=b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix}1 & 2 & 1 & 4\\ 2 & 4 & 2 & 8\\ 1 & 3 & 5 & 2\end{bmatrix}$. 1-qadam $(m_{21}=2, m_{31}=1)$: $R_2-2R_1=(0,0,0|0)$ — ikkinchi tenglama birinchisining ikki barobar katta; $R_3-R_1=(0,1,4|-2)$. $2va3$-qatorlarni almashtirgandan keyin: $\begin{bmatrix}1 & 2 & 1 & 4\\ 0 & 1 & 4 & -2\\ 0 & 0 & 0 & 0\end{bmatrix}$. Ozod o‘zgaruvchi $z=t$: $y=$

$-2-$ 4t, $x = 4-$ 2y $-z = 8+$ 7t. Cheksiz ko‘p yechimlar: $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix}$, $t \in R$; maxsus yechim $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ $(t = -1)$. Tekshirish: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$.

Mashq 32. 3×3 **chiqarib tashlash.** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$ **uchun** $Ax = b$ **ni Gauss chiqarib tashlash usuli bilan yeching.**

Yechim. Kengaytirilgan matritsa $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. 1-qadam $(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. 2-qadam $(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3 | -3)$. Orqaga almashtirish: $z = -1, y = 2, x = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

Mashq 33. $LUvaLDU3\times3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni hisoblang.**

Yechim. $m_{21} = 1, m_{31} = -1, m_{32} = 1$: $R_2 - R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$; $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; tekshirish: $LU = A$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 34. $LUvaLDU3\times3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni hisoblang.**

Yechim. $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 2$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$; $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; tekshirish: $LU = A$ (uchinchi qator: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U = [1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

Mashq 35. $LUvaLDU3\times3$**.** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$ **uchun** $LUvaLDU$ **ni hisoblang.**

Yechim. $m_{21}=1$, $m_{31}=-1$, $m_{32}=3$: $R_2-R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$; $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1&0&0\\1&1&0\\-1&3&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&4&1\\0&0&3\end{bmatrix}$; tekshirish: $LU=A$ (uchinchi qator: $-\begin{bmatrix}2&1&1\end{bmatrix}+3\cdot\begin{bmatrix}0&4&1\end{bmatrix}+\begin{bmatrix}0&0&3\end{bmatrix}=\begin{bmatrix}-2&11&5\end{bmatrix}$). $D=\begin{bmatrix}2&0&0\\0&4&0\\0&0&3\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/4; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 36. $LUvaLDU3\times3$**.** $A=\begin{bmatrix}2&1&1\\4&4&3\\-2&1&1\end{bmatrix}$ **uchun** $LUvaLDU$ **ni hisoblang.**

Yechim. $m_{21}=2, m_{31}=-1, m_{32}=1$: $R_2-2R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$; $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1&0&0\\2&1&0\\-1&1&1\end{bmatrix}$, $U=\begin{bmatrix}2&1&1\\0&2&1\\0&0&1\end{bmatrix}$; tekshirish: $LU=A$. $D=\begin{bmatrix}2&0&0\\0&2&0\\0&0&1\end{bmatrix}$, $D^{-1}U=[1\,\frac{1}{2}\,\frac{1}{2}$; 0 1 1/2; 0 0 1]; $A=LU=L\cdot D\cdot D^{-1}U$.

Mashq 37. Qatorlarni almashtirish va PLU**.** $A=\begin{bmatrix}0&2&1\\1&1&0\\2&3&2\end{bmatrix}$ **uchun qatorlarni almashtirish** $PvaPA=LU$ **tenglikni qanoatlantiruvchi omillarni toping.**

Yechim. $1va2$-qatorlarni almashtirish (birinchi asosiy element nol): $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$, $PA=\begin{bmatrix}1&1&0\\0&2&1\\2&3&2\end{bmatrix}$. Chiqarib tashlash: $m_{31}=2(R_3-2R_1=(0,\,1,\,2))$; $m_{32}=\frac{1}{2}(R_3-(\frac{1}{2})R_2=(0,\,0,\,\frac{3}{2}))$. $L=[100$; 0 1 0; $2\frac{1}{2}1], U=[110$; 0 2 1; 0 0 3/2]; tekshirish: $LU=PA$. Shunday qilib, $PA=LU$.

Mashq 38. Transponir va teskari. $A=\begin{bmatrix}1&2&0\\0&1&1\\2&0&1\end{bmatrix}$ **uchun** A^T **ni hisoblang va** $(A^T)^{-1}=(A^{-1})^T$ **tenglikni tekshiring.**

Yechim. $A^T=\begin{bmatrix}1&0&2\\2&1&0\\0&1&1\end{bmatrix}$; $\det A=5$. $(A^T)^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&2&-2\\-2&1&4\\2&-1&1\end{bmatrix}$; bosh tomondan, $A^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&-2&2\\2&1&-1\\-2&4&1\end{bmatrix}$, va $(A^{-1})^T=(\frac{1}{5})\cdot\begin{bmatrix}1&2&-2\\-2&1&4\\2&-1&1\end{bmatrix}=(A^T)^{-1}$. Tenglik bajarilgan; tekshirish: $A^T\cdot(A^T)^{-1}=I$.

Mashq 39. Simmetrik LDL^T. Simmetrik $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ ni LDL^T ko‘rinishida ajrating.

Yechim. Omillar $m_{21} = 1$, $m_{31} = 0$, $m_{32} = \frac{1}{3}$. $L = [100;\ 1\ 1\ 0;\ 0\frac{1}{3}1], D = [200;\ 0\ 3\ 0;\ 0\ 0\ 8/3]$, va $A = LDL^T$. Tekshirish: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

Mashq 40. Masala-muhokama: uch diagonallik modeli. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ **uchun $Ax = b$ ni LU bilan yeching. Eng yaqin qo‘shnilar tuzilishini talqin qiling.**

Yechim. LU: $L = [100;\ -1/2\ 1\ 0;\ 0 - \frac{2}{3}1], U = [2 - 10;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$ (omillar $m_{21} = -\frac{1}{2}$, $m_{32} = -\frac{2}{3}$ Lda asl belgilar bilan saqlanadi). To‘g‘ri almashtirish $Ly = by = [2;\ 1;\ 8/3]$ beradi; orqaga almashtirish $Ux = yx_3 = 2$, $x_2 = 2$, $x_1 = 2$ beradi. Shunday qilib, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Tekshirish: $A \cdot x = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. Matritsa uch diagonallik — har bir noma’lum faqat eng yaqin qo‘shnilar bilan bog‘langan; omillar L va U ikki diagonallik, va tor tuzilma yechimni arzon qiladi.

[Manba: 6-haftoq qo‘shimcha material — Misollar bilan qayta ko‘rib chiqish: I) LU, LDU va normalangan U qisqa darsi; II) 3×3 tizimlar uchun Gaussexclusion qisqa darsi; III) Gausss-Jordan usuli orqali teskari matritsani topishning batafsil misoli]

II darsning asosiy nuqtalari (5-hafto takrorlash). Tizim uchta tasvirda ko‘rib chiqilishi mumkin: qatorli (tenglamalar — tekisliklar), matritsali ($Ax = b$) va ustunli (A ustunlarining b ni berishi uchun kombinatsiyasi). Qatorlar uchun uchta aylantiriladigan amaliyonga ruxsat beriladi: ikki tenglamani almashtirish ($R_i \leftrightarrow R_j$), tenglamani noldan farqli songa ko‘paytirish ($R_i \leftarrow c \cdot R_i, c \neq 0$) va boshqa tenglamaning ko‘paytmasini qo‘shish ($R_i \leftarrow R_i + c \cdot R_j$); har biri butun tenglamaga, o‘ng qismni ham qatnashgan holda qo‘llaniladi. 3×3 tizim uchun protsedura: kengaytirilgan matritsani yozing; birinchi nol bo‘lmagan asosiy elementni tanlang (kerak bo‘lsa qatorlarni almashtiring); uning ostidagi elementlarni chiqaring; yangilangan matritsada ikkinchi asosiy elementni tanlang; uning ostidagi elementni chiqaring; ziddiyatni yoki erkin o‘zgaruvchilarni tekshiring; uchta asosiy element bo‘lganda teskiri almashtirishni amalga oshiring; yechimni asl tizimda tekshiring. Teskiri almashtirish formulalari: $z = c_3/u_{33}$, $y = (c_2 - u_{23} \cdot z)/u_{22}$, $x = (c_1 - u_{12} \cdot y - u_{13} \cdot z)/u_{11}$ (asosiy elementlar noldan farqli). Odatiy xatolar: o‘ng qismni yangilashni unutish; o‘zgarishidan keyin eski qatorni foydalanish; ko‘paytirchilar munosabati yo‘nalishini chalkashtirish; taklif qilingan nol asosiy elementni degeneratsiya isboti deb hisoblash.

Misol 1. LU**, normalash va** LDU **(dars** I**).** $A = \begin{bmatrix} 2 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{bmatrix}$ **matritsasi** $LUvaLDU$ **ajratmalarini toping.**

Yechim. Birinchi asosiy element 2 ostidagi elementlarni chiqaramiz: $R_2 \leftarrow R_2 - 2R_1 va R_3 \leftarrow R_3 - R_1 \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 6 \end{bmatrix}$ beradi; keyin ikkinchi asosiy element 3 ostida: $R_3 \leftarrow R_3 - R_2 \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$ beradi. Ko‘paytchilar $m_{21} = 2$, $m_{31} = 1$, $m_{32} = 1$, shuning uchun $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ $va A = LUU = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$ bilan; L asl belgilar bilan ko‘paytchilarni saqlaydi (ayirish amaliyati $2R_1$ ni ayiradi, lekin $L_{21} = +2$). Normalash: har bir U qatorini uning diagonal elementiga bo‘lamiz — asosiy elementlar $2,\ 3,\ 3 D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ ga chiqariladi, normalangan yuqori ko‘paytma $D^{-1}U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, va $A = L \cdot D \cdot (D^{-1}U)$; D^{-1} ga chapdan ko‘paytirish qatorlarni masshtablaydi. E’tibor berish: $A = LU$ da asosiy elementlar U diagonalida joylashgan (D ajratilmaydi, birlik diagonal — Lda); $A = LDU$ da asosiy elementlar Dda, birlik diagonallar esa L va uchinchi ko‘paytmada; ko‘p darsliklar $D^{-1}UniU$ deb qayta belgilaydi.

Misol 2. Faktorlar bo‘yicha tizimni yechish (dars I**). Misol 1 ajratmasidan foydalanib,** $Ax = \begin{bmatrix} 8 \\ 22 \\ 17 \end{bmatrix}$ **ni yeching.**

Yechim. To‘g‘ri almashtirish $Ly = b$: $y_1 = 8$; $2y_1 + y_2 = 22$, bundan $y_2 = 6$; $y_1 + y_2 + y_3 = 17$, bundan $y_3 = 3$, ya’ni $y = \begin{bmatrix} 8 \\ 6 \\ 3 \end{bmatrix}$. Teskiri almashtirish $Ux = y$: $3x_3 = 3, x_3 = 1$; $3x_2 + 3x_3 = 6, x_2 = 1$; $2x_1 + 4x_2 + 2x_3 = 8$, $x_1 = 1$. Shunday qilib, $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. LDU bilan hisoblash uch bosqamga bo‘linadi: $Ly = b, Dz = y$ (asosiy elementlarga bo‘lish), keyin $(D^{-1}U)x = z$. Qatorlarni almashtirish kerak bo‘lganda, standart shakl — $PA = LU$ yoki $PA = LDU$, bu erda P almashtirishlarni yozadi; asosiy elementlarni chiqarish va bo‘lish ularning noldan farqliligini talab qiladi.

Misol 3. Gaussexlectionning to‘liq misoli (dars II**).** $\mathbf{2x} + y - z = 1$**,** $\mathbf{4x} + \mathbf{3y} + z = 13, -\mathbf{2x} + y + \mathbf{2z} = 6$ **tizimini yeching.**

Yechim. Bu erda $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}$. Bosqich 1: birinchi asosiy element 2, ko‘paytchi $m_{21} = \frac{4}{2} = 2$; $R_2 \leftarrow R_2 - 2R_1$: $(4x + 3y + z) - 2(2x + y - z) = 13 - 2 \cdot 1$, ya’ni $y+$ 3z $= 11$. Bosqich 2: $m_{31} = -\frac{2}{2} = -1$; $R_3 \leftarrow R_3 + R_1$: $(-2x + y + 2z) + (2x + y - z) = 6 + 1$, ya’ni 2y $+z = 7$. Bosqich 3: ikkinchi asosiy element 1 (yangilangan 2-tenglamadagi y koeffitsienti), $m_{32} = 2$; $R_3 \leftarrow R_3 - 2R_2$: $(2y + z) - 2(y + 3z) = 7 - 22$, ya’ni $-$5z $= -15$. Asosiy elementlar 2, 1, -5 — ikkinchi va uchinchisi yangilangan tenglamalardan, asl diagonal emasdan olinadi. Teskiri almashtirish: $z = 3, y = 2, x = 1$, ya’ni $(x, y, z) = (1, 2, 3)$. Tekshirish: $2 \cdot 1 + 2 - 3 = 1$; $4 \cdot 1 + 6 + 3 = 13$; $-2 \cdot 1 + 2 + 6 = 6$. Kengaytirilgan matritsa bilan: $\begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 3 & 1 & 13 \\ -2 & 1 & 2 & 6 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 2 & 1 & 7 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & -5 & -15 \end{bmatrix}$.

Misol 4. Exclusionni matritsa ko‘paytirishi sifatida (dars II). Misol 3 amaliyotlarini exclusion matritsalari sifatida yozing.

Yechim. $R_2 \leftarrow R_2 - 2R_1$ uchun: $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ($E_{21} \cdot A$ ko‘paytmaning ikkinchi qatori $-2 \cdot (A$ ning 1-qatori$) + (A$ ning 2-qatriga) teng); 1-qatorni 3-qatorga qo‘shish uchun: $E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; $R_3 \leftarrow R_3 - 2R_2$ uchun: $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. To‘liq chiqarib tashlash: $E_{32} \cdot E_{31} \cdot E_{21} \cdot A = Uva E_{32} \cdot E_{31} \cdot E_{21} \cdot b = c$, bunda $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 11 \\ -15 \end{bmatrix}$; eng o‘ngdagi operatsiya birinchi ta’sir ko‘rsatadi. Qoidalar: $(BC)_{ij} = \sum_k b_{ik} \cdot c_{kj}$; assotsiativlik $(BC)D = B(CD)$; kommutativsizlik $BC \neq CB$ — shuning uchun qator operatsiyalari tartibi muhim.

5-misol. Nol taklif etilgan asosiy element (II dars). $y + z = 5, x + y + z = 6$, 2x + 3y + 4z $= 20$ tenglamalar sistemasini yeching.

Yechim. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 6 \\ 20 \end{bmatrix}$; birinchi taklif etilgan asosiy element nol — $1va2$-qatorlarni almashtiramiz (almashtirish matritsasi $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, qayta tartiblangan sistema PAx $= Pb$): $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 2 & 3 & 4 & 20 \end{bmatrix}$. Keyin $R_3 \leftarrow R_3 - 2R_1$: (0, 1, 2 | 8); keyin $R_3 \leftarrow R_3 - R_2$: (0, 0, 1 | 3). Orqaga almashtirish: $z = 3, y = 2, x = 1$. Taklif etilgan asosiy pozitsiyadagi nol avtomatik ravishda yagona yechimning yo‘qligini anglatmaydi; sonli hisoblar uchun qisman tanlov joriy ustunning modul bo‘yicha eng katta mavjud elementni oladi, chiqarib tashlash ko‘paytmalarini nazorat qiladi.

6-misol. Sistemaning nechta yechimi bor (II dars). $x + y + z = 6$, 2x + 3y + 4z

$= 20$**, 3x + 4y + 5z** $= k$ **oilasini ko‘rib chiqing. qanday** k **uchun yechimlar yo‘q, qanday** k **uchun cheksiz ko‘p?**

Yechim. $R_2 \leftarrow R_2 - 2R_1 va R_3 \leftarrow R_3 - 3R_1$ qo‘llaymiz: $y+$ 2z $= 8 vay+$ 2z $= k - 18$ olinadi. 2-tenglamani 3-tenglamadan ayiramiz: $0 = k - 26$. Agar $k \neq 26$ — yechimlar yo‘q ($k = 27$ bo‘lganda $0 = 1$ — tenglamalar mos emas). Agar $k = 26$ — oxirgi tenglama $0 = 0$ bo‘lib, cheklov qo‘shmaydi: erkin o‘zgaruvchi $z = t$ uchun $y = 8-$ 2t va $x = 6 - y - z = t - 2$, ya’ni $(x,\, y,\, z) = (t - 2,\, 8 - 2t,\, t)$, $t \in R$ — cheksiz ko‘p yechimlar. 3×3 uchun natijalar jadvali: uchta asosiy element — yagona yechim; $[000|d]$ qatori, $d \neq 0$ — yechimlar yo‘q; uchta asosiy elementdan kam, ziddiyatli qator yo‘q — cheksiz ko‘p yechimlar (bitta nol qati o‘z-o‘zidan moslikni kafolatlamaydi — boshqa qatorlarni tekshirish kerak).

7-misol. LU**-faktorizatsiya va qayta foydalanish (II dars). 3-misol sistemasining** $A = LU$ **ni yozing va** $Lc = b, Ux = c$ **ni yeching.**

Yechim. 3-misol ko‘paytmalari: $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 2$; ularni diagonal ostiga joylaymiz: $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, va $A = LUU = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$ bilan; L dagi belgilar o‘zlari ko‘paytmalardir, chiqarib tashlash operatsiyalari ularning manfiy qiymatlaridan foydalanadi. Ikkita bosqichda yechamiz: $Lc = b c_1 = 1$, $c_2 = 11$, $c_3 = -15$ beradi (to‘g‘ri almashtirish); keyin $Ux = c$ orqaga almashtirish orqali (1, 2, 3) olinadi. Afzallik bir nechta sistemalarda bir A matritsasi, lekin turli o‘ng tomonlar bo‘lganda namoyon bo‘ladi: A bir marta ajratiladi, $LvaU$ qayta foydalaniladi. Agar qator almashtirishlar kerak bo‘lsa, standart faktorizatsiya — $PA = LU$.

8-misol. $AvaB$ **mashqlari (II dars). Sistemalarni yeching:** $(A) x + y + z = 6$**, 2x +** $3y + z = 11, x +$ **2y + 3z** $= 14$**;** $(B) y +$ **2z** $= 8, x + y + z = 6$**, 2x + 3y** $+ z = 11$**.**

Yechim. $(A) x$ ni chiqaramiz: $R_2 - 2R_1 y - z = -1$ beradi; $R_3 - R_1 y +$ 2z $= 8$ beradi. Ayirish: 3z $= 9$, $z = 3$, $y = z - 1 = 2$, $x = 6 - 2 - 3 = 1$. Javob (1, 2, 3); tekshirish: $1 + 2 + 3 = 6$, $2 + 6 + 3 = 11$, $1 + 4 + 9 = 14$. (B) Birinchi tenglamada x yo‘q (nol taklif etilgan asosiy element) — x ni chiqarishdan oldin birinchi ikki qatorni almashtiramiz: $x + y + z = 6, y +$ 2z $= 8$, 2x + 3y $+ z = 11$. x ni chiqaramiz: $R_3 - 2R_1 y - z = -1$ beradi; $y +$ 2z $= 8$ bilan birga ayirish 3z $= 9, z = 3, y = 2, x = 1$ beradi. Javob (1, 2, 3); tekshirish: $2 + 2 \cdot 3 = 8$, $1 + 2 + 3 = 6$, $2 + 6 + 3 = 11$.

9-misol. $CvaD$ **mashqlari (II dars).** (C) **qanday** k **uchun** $x + y + z = 6$**, 2x + 3y + 4z** $= 20$**, 3x + 4y + 5z** $= k$ **sistemasida yechimlar bo‘lmaydi, qanday** k **uchun cheksiz ko‘p yechimlar borligini aniqlang.** $(D) R_3 \leftarrow R_3 - 4R_1$ **operatsiyasining matritsasi topiling.**

Yechim. $(C) Bu 6$-misol oilasi: uchinchi tenglamani chiqarib tashlagandan so‘ng $u0 = k - 26$ ga aylanadi, shuning uchun $k \neq 26$ bo‘lganda yechim yo‘q, $k = 26$ bo‘lganda cheksiz ko‘p yechim mavjud: $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$; yagona yechim beradigan k qiymati mavjud emas (koeffitsientlar matritsida faqat ikkita asosiy element bor). (D) Amaliyot faqat 3-qatorni o‘zgartiradi, undan $4 \cdot (1-$qator) ayirilib: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; $E \cdot A$ ko‘paytmasining uchinchi qatori (A ning 3-qatori) $- 4 \cdot (A$ ning 1-qatori) ga teng, qolgan qatorlar o‘zgarmaydi.

10-misol. Gauss-Jordan usuli bilan teskari matritsa (III dars). $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

matritsasining teskari matritsini toping, birlik matritsa qo‘shib va chap to‘liq birlik matritsaga keltiring.

Yechim. Nazariya: n×n A matritsasining teskarisi A^{-1} matritsasi bo‘lib, $AA^{-1} = A^{-1}A = I$; qatorlardagi har bir elementar amaliyot chapdan elementar matritsaga ko‘paytirishga teng, va agar $E_k \cdots E_2E_1A = I$ bo‘lsa, u holda $A^{-1} = E_k \cdots E_2E_1$ — shuning uchun I ga qo‘llanadigan xuddi shu amaliyotlar A^{-1} ni beradi: $[A|I] \to [I|A^{-1}]$; har bir amaliyot butun kengaytirilgan qatorga ta’sir qiladi. 1-qadam: $[A|I] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. 2-qadam (birinchi asosiy element ostini chiqarish): $R_2 \leftarrow R_2 - R_1$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. 3-qadam (ikkinchi ostida): $R_3 \leftarrow R_3 - R_2$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. 4-qadam (uchinchisida ustida): $R_2 \leftarrow R_2 - R_3$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. 5-qadam (ikkinchisida ustida): $R_1 \leftarrow R_1 - R_2$: $\begin{bmatrix} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Chap to‘liq birlik matritsaga aylandi, shuning uchun $A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. Ko‘paytirish orqali tekshirish: $A \cdot A^{-1} = I$ (masalan, (2, 1) elementi: $1 \cdot 3 + 2 \cdot (-2) + 1 \cdot 1 = 0$).

11-misol. Qachon usul ishlamaydi (III dars). $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ matritsasining teskarisi yo‘qligini ko‘rsating.

Yechim. Kvadrat matritsa faqat $[A|I]$ dagi chap blok birlik matritsaga keltirilgagina teskariga ega. $R_2 \leftarrow R_2 - 2R_1$ ni qo‘llaymiz: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix}$. Chap blokda nol qator paydo bo‘ldi — I ni hosil qilish uchun yetarlicha asosiy elementlar yo‘q; matritsa singulyar (notekis), teskarisi mavjud emas. Eslatma: Gauss chiqarib tashlash yuqori uchburchak matritsada to‘xtaydi, Gauss-Jordan esa har bir asosiy element $1ga$, va har bir asosiy element ustidagi va ostidagi barcha elementlar nolga aylanguncha davom ettiradi.

4 7-hafta — Transponirlash, almashtirishlar va vektorli fazolar (Chiziqli algebra I)

[Manba: 7-hafta ma’ruzasi ”Transponirlar, almashtirishlar, vektorli fazolar va kichik fazolar” (Strang, Introduction to Linear

Algebra, 4th ed., §2.7 pp. 108–120, §3.1 pp. 121–132)]

4.5.1 6-haftani takrorlash: $A = LU, LDU, PA = LU$

Gauss chiqarishi $E_k \cdots E_1A = U$ beradi. Qatorlarni almashtirmasdan, chiqarish ko‘paytmalari pastburchakli L (bosh diagonalda birlar) to‘planadi va $A = LU$: masalan,

$A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$, $R2 \leftarrow R2 - 2R1$ bosqichi $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ ni beradi. Bosh elementlarni ajratib, $A =$ LDŕ hosil qilamiz: $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ — bosh elementlardan diagonal, Ū (U chiziq bilan) $= [1\frac{1}{2}; 0\ 1]$ — bosh diagonalda birlar bilan yuqori burchakli.

$$A = LU, A = \text{LD}\bar{\text{U}}; almashtirishlaruchun — PA = LU$$

Agar bosh element nol bo‘lsa ($A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$), qatorlar almashtiriladi: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$ ni beradi; barcha qator almashtirishlari P da yoziladi.

4.5.2 Transponirlash: ta’rif va asosiy identifikatlar

Agar A — $m \times n$ matritsa bo‘lsa, A^T o‘lchami $n \times m$ bo‘ladi va elementlari $(A^T)_{ij} = a_{ji}$ bo‘ladi: ustunlar satrlarga aylanadi. Misol: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ ni beradi.

$$(A^T)^T = A, (A + B)^T = A^T + B^T, (cA)^T = cA^T$$

Element bo‘yicha isbot (yig‘indiga uchun): $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$.

$$(AB)^T = B^T A^T$$

Ko‘paytuvchilar tartibi teskarisiga o‘zgaradi (A^{}{T} B^{}{T} hatto aniqlanmagan bo‘lishi mumkin). Isbot: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Dars tekshiruvi: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$: $(AB)^T = \begin{bmatrix} 8 & 3 \\ 2 & 1 \end{bmatrix} = B^T A^T$.

$$(A^{-1})^T = (A^T)^{-1}$$

Invertirilgan A uchun transponir qilamiz $AA^{-1} = I$: $(A^{-1})^T A^T = I$ ni olamiz; xuddi shunday $A^T (A^{-1})^T = I$. Demak, transponir va teskari amallar o‘zaro almashtiriladi.

4.5.3 Simmetrik va antimetrik matritsalar; $A^T A$ ko‘paytmasi

Matritsa S simmetrik, agar $S^T = S (s_{ij} = s_{ji})$; misol: $S = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$. Matritsa K antimetrik, agar $K^T = -K$; misol: $K = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$; asosiy diagonal nolli: $k_{ii} = -k_{ii} \Rightarrow k_{ii} = 0$.

$$A = (A + A^T)/2 + (A - A^T)/2 = S + K, S^T = S, K^T = -K$$

Ixtiyoriy kvadrat matritsa shunday ajraladi; dars misoli: $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \rightarrow S = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; tekshirish: $S + K = A$.

$$(A^T A)^T = A^T A, x^T A^T A x = (Ax)^T (Ax) = ||Ax||^2 \geq 0$$

A^TA ko‘paytmasi har doim simmetrikdir ($(AB)^T = B^TA^T$ maxsus holati), va $x^TA^TAx = ||Ax||^2 \geq 0$ har qanday x uchun. Misol: $A = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ $A^TA = \begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}$ ni beradi; R^TR matritsalar — eng kichik kvadratlar asosi. Skalyar va tashqi ko‘paytma: ustunlar $u, v \in R^n$ uchun u^Tv soni skalyar ko‘paytma, uv$^T esan \times n$ matritsadir. Misol: $u = \begin{bmatrix}1\\ 2\end{bmatrix}$, $v = \begin{bmatrix}3\\ 4\end{bmatrix}$: $u^Tv = 11$, uv$^T = \begin{bmatrix}3 & 4\\ 6 & 8\end{bmatrix}$.

4.5.4 Perestanovkalar: $P^T = P^{-1}$, oltita 3×3 matritsa, qisman elementlar almashtirish

Perestanovka matritsi birlik matritsaning satrlari almashtirilishidan hosil bo‘ladi: har bir satr va ustunda aynan bitta birlik bo‘ladi. Misol: $P = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ $[x_1;\, x_2;\, x_3]ni[x_2;\, x_3;\, x_1]$ ga o‘tkazadi.

$$PA—matritsa Abilan almashtirilgan qatorlar;\; AQ—almashtirilgan ustunlar bilan$$

Chapdan ko‘paytirish satrlarni almashtiradi: $A = \begin{bmatrix}1 & 2\\ 3 & 4\\ 5 & 6\end{bmatrix}$, $P = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ $\rightarrow$ $PA = \begin{bmatrix}3 & 4\\ 5 & 6\\ 1 & 2\end{bmatrix}$. O‘ngdan ko‘paytirish ustunlarni almashtiradi: $A = \begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$, $Q = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$ $\rightarrow$ $AQ = \begin{bmatrix}2 & 3 & 1\\ 5 & 6 & 4\end{bmatrix}$. Umuman olganda $PA \neq AP$.

$$P^TP = I, ya'ni P^{-1} = P^T;\; kelib chiqishi: ||Px|| = ||x||, \det P = \pm 1$$

P ustunlari — turli koordinata birlik vektorlari, shuning uchun ularning skalyar ko'paytmalari I ni hosil qiladi: $P^TP = I$, bundan $P^{-1} = P^T$. Natijalar: $\|Px\| = \|x\|$ va $\det P = \pm 1$. Bir qatorni almashtirish determinant belgisini o'zgartiradi: $\det \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = -1$, shuning uchun $\det(PA) = -\det A$; ikki almashtirish belgisini qaytaradi. n tartibli almashtirishlar soni aynan $n!$; $n = 3$ bo'lganda oltitasi bor (Strang, §2.7) — uchta juft ($\det P = +1$) va uchta toq ($\det P = -1$); $P^{-1} = P^T$, shuning uchun $(P_{32}P_{21})^{-1} = P_{21}P_{32}$:

Almashtirish	**Matritsa (qator bo‘yicha)**	$\det P$	$P^{-1} = P^T$
I	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	$+1$	I
P_{21} (1, 2-qatorlarni almashtirish)	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	-1	P_{21}
P_{32} (2, 3-qatorlarni almashtirish)	$\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$	-1	P_{32}
P_{31} (1, 3-qatorlarni almashtirish)	$\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$	-1	P_{31}
$P_{32}P_{21}$ (tsiklik)	$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	$+1$	$P_{21}P_{32}$
$P_{21}P_{32}$ (tsiklik)	$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	$+1$	$P_{32}P_{21}$

Almashtirish	Matritsa (qator bo‘yicha)	$\det P$	$P^{-1} = P^T$

$P32P21$ (tsiklik) $\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix} + 1P21P32P21P32$(tsiklik) $\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix} + 1P32P21$

Asosiy elementning qisman tanlovi (partial pivoting): agar asosiy element nol bo‘lsa, qatorlar almashtiriladi, joriy ustundagi moduli eng katta bo‘lgan nomzod tanlanadi. Misol: $A = [10^{-5}1;\ 1\ 1]$: almashtirishsiz ko‘paytuvchi $\frac{1}{10}^{-5} = 10^5$, almashtirishdan keyin — $10^{-5}/1 = 10^{-5}$. Almashtirishlar P da yoziladi: $PA = LU$.

4.5.5 Vektorli makonlar: ta’rif va misollar

Vektorli makon R ustida — vektorlar to‘plami qo‘shish va skalyar ko‘paytirish amaliyotlari bilan, vektorli makon aksiomalatiga javob beradi: nol vektor, qarama-qarshi elementlar, assotsiativlik, distributivlik va $h.k.$ Dars misollari: R^n, haqiqiy matritsalar, ko‘phadlar, uzluksiz funksiyalar — ”vektor” har qanday ob’ekt bo‘lishi mumkin.

4.5.6 Alt makonlar va yopilishni tekshirish

$$W \subseteq V — kichik fazolar \Leftrightarrow 0 \in W; u + v \in W barchau, v \in W uchun; cu \in W barchau \in W,$$
$$c \in R uchun$$

Teng ravishda: bo‘sh bo‘lmagan $Wau+$ bv kombinatsiyalariga nisbatan yopiq. Tekshirish: (1) nol vektor; (2) yig‘indi; (3) skalyar ko‘paytirish — har qanday bandning buzilishi inkor etadi.

- Boshdan o‘tuvchi to‘g‘ri chiziq: $W = \{(t,\ 2t)\} = \text{span}\{(1,\ 2)\}$: $u = (s,\ 2s), v$

$=(t,$ 2t) uchun au $+bv=(as+bt,2(as+bt))$ yana W da — alt makon.

- Siljigan to‘g‘ri chiziq (kontremisol): $S=\{(x,y)$: $y=$ 2x $+1\}$ — nol vektor

tenglamaga javob bermaydi; alt makon koordinata boshidan o‘tishi shart.

- Boshlang‘ich nuqtadan o‘tuvchi tekislik: $W=\{(x,y,z)$: $x+y+z=0\}=$

span{(1, 0, −1), (0, 1, −1)} $(z=-x-ydan:(x,$ $y,$ $z)=x(1,0,-1)+y(0,1,-1))$; $x+y+z=1$ tekisligi — kichik fazo emas (nol yo‘q).

- R^3 geometriyasi: kichik fazolar — faqat {0}, 0 dan o‘tuvchi to‘g‘ri chiziqlar, 0

dan o‘tuvchi tekisliklar va butun R^3.

- Kesishish va birlashma: $U\cap W$ — kichik fazo; birlashma — albatta emas: x

va y o‘qlari R^2 ning kichik fazolari, lekin $(1,0)+(0,1)=(1,1)$ hech qanday oqda joylashmagan.

- Matritsalar va ko‘phadlar: simmetrik 2×2 — barcha 2×2 matritsalar kichik

fazosi (yig‘indilar va ko‘paytmalar simmetrik); $p(0)=0$ bo‘lgan ko‘phadlar — kichik fazo, $p(0)=1$ bo‘lganlari — yo‘q.

4.5.7 Chiziqli birikmalar va chiziqli qoplamalar (span)

$$\text{span}v_1,...,v_k=c_1v_1+...+c_kv_k:c_i\in R$$

Chiziqli qoplama — berilgan vektorlarning barcha birikmalarining to‘plami; u har doim kichik fazodir: agar $u=\Sigma a_iv_ivav=\Sigma b_iv_i$ bo‘lsa, su $+tv=\Sigma(sa_i+tb_i)v_i$ — yana bir birikma. Dars misoli: $(3,5)$span{(1, 1), (1, −1)} da joylashganmi? $c_1+c_2=3$, $c_1-c_2=5$ tenglamalarini yechamiz $\to c_1=4,c_2=-1$ — ha.

4.5.8 Ustunlar fazosi $C(A)$

$$A=[a_1a_2...a_n]\to Ax=x_1a_1+...+x_na_n;$$
$$C(A)=Col(A)=\text{span}a_1,...,a_n$$

Ax ko‘paytmasi — A ustunlarining chiziqli birikmasi. Ustunlar fazosi $C(A)=\text{Col}(A)$ — ustunlarning chiziqli qoplamasi, $m\times n$ matritsa A uchun R^m ning kichik fazosi. Misol: $A=\begin{bmatrix}1&2\\2&4\\0&0\end{bmatrix}$ — ustunlar proporsional, $\text{Col}(A)=$ span{(1, 2, 0)} — R^3 dagi to‘g‘ri chiziq.

$$tizimAx=byechimgaega\Leftrightarrow b\in C(A)$$

$Ax=b$ tizimida yechim bo‘lishi shart va shartli ravishda b ustunlar fazosida joylashgan bo‘lishi bilan bog‘liq: yechim xb ni beradigan ustunlar birikmasining koeffitsientlarini tanlaydi. Yechim mavjudligining asosiy mezonini bilish kerak.

4.5.9 Nol fazosi $N(A)$

$$N(A) = Null(A) = x : Ax = 0$$

Nol fazosi — bir xil tizimning yechimlari to‘plami; bu R^n ning kichik fazosi: agar $Au = Av = 0$ bo‘lsa, $A(su + tv) = 0$. Misol: $A = \begin{bmatrix}1 & 2\end{bmatrix}$: $x_1 + 2x_2 = 0 x_1 = -2x_2$ beradi, shuning uchun Null$(A) =$ span$\{(-2, 1)\}$ — to‘g‘ri chiziq.

$$x : Ax = 0 — hardoimkichikfazolar; x : Ax = b,$$
$$b \neq 0 — kichikfazolaremas(0nio'zichigaolmaydi)$$

Bir xil tizimning yechimlari kichik fazoni hosil qiladi; bir xil bo‘lmagan $(b \neq 0)$ tizimning yechimlari — yo‘q: nol vektor ularga tegishli emas. Agar $Ax = b$ tizimi yechimga ega bo‘lsa, uning yechimlari nol fazosining siljishi: $x = x_p + x_n$, bunda x_p — maxsus yechim, $x_n \in$ Null(A) (affin to‘plami). 8-hafta o‘rtasiga qadar: $A^T va (AB)^T$ ni hisoblash, $P^{-1} = P^T va PA = LU$ dan foydalanish, yopiq holatni tekshirish, qoplamalarni va tegishlikni $C(A)$ topish.

[Manba: 7-hafta dars mashqlari (Yechilgan), Chiziqli algebra $I(Kuz2026)$ — 20 dars mashqlasi; arifmetika pythonda
tekshirilgan]

Mashq 1. 2×2 **matritsani transponirovka qilish.** $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ **uchun** A^T **ni hisoblang.**

Yechim. Transponirovka qatorlar va ustunlarni almashtiradi: $(A^T)_{ij} = a_{ji}$. Shuning uchun $A^T = \begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$; ayniqsa, $(A^T)_{12} = a_{21} = 3$, diagonal o‘zgarmaydi.

Mashq 2. Yig‘indining transponirovkasi. $A = \begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}, B = \begin{bmatrix}2 & 1\\ 0 & -1\end{bmatrix}$ **uchun** $(A+B)^T$ **ni hisoblang.**

Yechim. Avval qo‘shamiz: $A + B = \begin{bmatrix}3 & 1\\ 2 & 2\end{bmatrix}$. Keyin transponirovka qilamiz: $(A + B)^T = \begin{bmatrix}3 & 2\\ 1 & 2\end{bmatrix}$. Tekshirish: $A^T + B^T = \begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix} + \begin{bmatrix}2 & 0\\ 1 & -1\end{bmatrix} = \begin{bmatrix}3 & 2\\ 1 & 2\end{bmatrix}$ — mos keldi.

Mashq 3. $A^T A$ **ko‘paytmasi.** $A = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ **uchun** $A^T A$ **ni toping.**

Yechim. $A^T A = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} \cdot \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}$ — natija simmetrik, har doimgidek (ISA 26-mashq).

Mashq 4. Ko‘paytmaning transponirovkasi. $A = \begin{bmatrix}1 & 1\\ 0 & 2\end{bmatrix}, B = \begin{bmatrix}2 & 0\\ 1 & 3\end{bmatrix}$ **uchun** $(AB)^T$ **ni toping.**

Yechim. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, shuning uchun $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$. Tenglik xuddi shuni beradi: $B^T A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$ — ko‘paytuvchilar teskari tartibda.

Mashq 5. Simmetriklikni tekshirish. $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ **matritsasi simmetrikmi?**

Yechim. $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$. Ha, simmetrik: mos keladigan diagonal tashqari elementlar bir xil $(a_{12} = a_{21} = -3)$, diagonal transponirovlanganda o‘zgarmaydi.

Mashq 6. Simmetrik va antimetrik qismlar. $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ **ni simmetrik va antimetrik qismlarga ajrating.**

Yechim. $A^T = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$ (diagonal tashqari elementlar o‘rtacha), $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Tekshirish: $S + K = A, S^T = S, K^T = -K$.

Mashq 7. Qatorlarni almashtirish: PA**.** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** PA **ni toping.**

Yechim. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$: ko‘paytmaning birinchi qatori A ning ikkinchi qatori va aksincha. Chapdan ko‘paytirish $1va2$-qatorlarni almashtiradi.

Mashq 8. Ustunlarni almashtirish: AP**.** $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **uchun** AP **ni toping.**

Yechim. $AP = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$: ko‘paytmaning birinchi ustuni A ning ikkinchi ustuni, va aksincha. O‘ngdan ko‘paytirish $1va2$ ustunlarini almashtiradi; shu bilan birga $AP \neq PA$.

Masala 9. Permutatsiya matritsasining teskari matritsasi. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **uchun** P^{-1} **ni toping.**

Yechim. Permutatsiya matritsasi uchun $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Tekshirish: $P^T P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = I$ — P ning ustunlariortonormalangan.

Masala 10. $1va3$**-qatorlarni almashtirish.** 3×3 **matritsaning** $1va3$**-qatorlarini almashtiruvchi permutatsiya matritsasi** P **ni toping.**

Yechim. $1va3$-qatorlarni almashtirish, I_3 ga qo‘llanilganda: $P=\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$. Har qanday A uchun ko‘paytma PA uning birinchi va uchinchi qatorlarini almashtiradi; masalan, $A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$ uchun $\begin{bmatrix}7&8&9\\4&5&6\\1&2&3\end{bmatrix}$ hosil bo‘ladi.

Masala 11. $PA=LU$ **nol bosh elementi bilan.** $A=\begin{bmatrix}0&2\\3&4\end{bmatrix}$ **uchun** P,L,U **ni toping.**

Yechim. Bosh element $a_{11}=0$ — almashtirmasdan bo‘lmaydi. Qatorlarni almashtiramiz: $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$, $PA=\begin{bmatrix}3&4\\0&2\end{bmatrix}$. Bosh element 3 ostida 0 — chiqarish tugadi, ko‘paytchilar yo‘q: $L=I_2$, $U=PA=\begin{bmatrix}3&4\\0&2\end{bmatrix}$; natija: $PA=LU$.

Masala 12. LU **qatorlarni almashtirmasdan.** $A=\begin{bmatrix}2&1\\6&5\end{bmatrix}$ **uchun** L,U **ni qatorlarni almashtirmasdan toping.**

Yechim. Bosh element $2\neq 0$, almashtirish kerak emas; ko‘paytchi $m_{21}=\frac{6}{2}=3$. Qadam $R2\leftarrow R2-3R1$: $\begin{bmatrix}6&5\end{bmatrix}-3\cdot\begin{bmatrix}2&1\end{bmatrix}=\begin{bmatrix}0&2\end{bmatrix}$, shuning uchun $U=\begin{bmatrix}2&1\\0&2\end{bmatrix}$, $L=\begin{bmatrix}1&0\\3&1\end{bmatrix}$. Tekshirish: $LU=\begin{bmatrix}2&1\\6&5\end{bmatrix}=A$.

Masala 13. Altbo‘shlik: boshdan o‘tuvchi to‘g‘ri chiziq. $W=$ **{**(x,y)**:** $y=$ **3x}** R^2 **ning altbo‘shligimi?**

Yechim. Ha. $W=\{t(1,3)\colon\ t\in R\}$: nol vektort $=0ga$ mos keladi. $u=(s,$ 3s$),v=(t,$ 3t$)va$ har qanday a,b uchun: au $+bv=(as+bt,3(as+bt))\in W$ — yopiqlik mavjud; W — altbo‘shlik (boshdan o‘tuvchi to‘g‘ri chiziq).

Masala 14. Misol: siljigan to‘g‘ri chiziq. $S=$ **{**(x,y)**:** $y=$ **3x** $+2$**} altbo‘shlikmi?**

Yechim. Yo‘q. Nol vektor tenglamani qanoatlantirmaydi: $3\cdot 0+2=2\neq 0$. Nol vektor testi inkor uchun yetarli: altbo‘shlik koordinata boshini o‘z ichiga olishi shart.

Masala 15. Altbo‘shlik: tekislik. $W=$ **{**(x,y,z)**:** $x+y-z=0$**} altbo‘shlikmi?**

Yechim. Ha. $f(x,\ y,\ z)=x+y-zvau$, $v\in W$ uchun: $f(au+bv)=a\cdot f(u)+b\cdot f(v)=0$, demak au $+bv\in W$; va $f(0,0,0)=0$. W — altbo‘shlik (boshdan o‘tuvchi tekislik).

Masala 16. Vektorlarning chiziqli birikmasi. $(4,\ 1)ni(1,\ 1)va(1,\ -1)$ **vektorlarining birikmasi sifatida ifodalang.**

Yechim. $c_1(1,\ 1)+c_2(1,\ -1)=(4,\ 1)$ ni qidiramiz: $c_1+c_2=4$, $c_1-c_2=1$. Qo‘shamiz: $2c_1=5$, $c_1=\frac{5}{2}$, $c_2=\frac{3}{2}$. Tekshirish: $(\frac{5}{2}+\frac{3}{2},\ \frac{5}{2}-\frac{3}{2})=(4,\ 1)$.

Masala 17. Proporsional vektorlarning qobig‘i. $(1,\ 2,\ 0)va(2,\ 4,\ 0)$ **vektorlarining chiziqli qobig‘ini toping.**

Yechim. Ikkinchi vektor birinchisining ikki barobar katta: $(2,4,0)=2(1,2,0)$, shuning uchun har qanday birlik (1, 2, 0) ga karrali. Natija: span{(1, 2, 0), (2, 4, 0)} = $\{t(1,2,0)\colon\ t\in R\}$ — R^3 dagi to‘g‘ri chiziq.

Masala 18. 1×3 **matritsaning nol bo‘shlig‘i.** $A=\begin{bmatrix}1 & 2 & -1\end{bmatrix}$ **uchun Null**(A) **ni toping.**

Yechim. $Ax=0$ tenglama: $x_1+2x_2-x_3=0$. Erkin o‘zgaruvchilar $x_2=s, x_3=t$, unda $x_1=-2\mathrm{s}+tvax=s(-2,\,1,\,0)+t(1,\,0,\,1)$. Natija: Null$(A)$ = span{(−2, 1, 0), (1, 0, 1)} — R^3 dagi tekislik. Tekshirish: $1\cdot(-2)+2\cdot 1-0=0va1\cdot 1+2\cdot 0-1=0$.

Masala 19. Ustunlar bo‘shlig‘i. $A=\begin{bmatrix}1 & 2\\ 2 & 4\\ 0 & 0\end{bmatrix}$ **uchun Col**(A) **ni toping; unga (3, 6, 0) vektori tegishlimi?**

Yechim. Ustunlar $a_1=(1,\,2,\,0)vaa_2=(2,\,4,\,0)=2a_1$ proporsional, shuning uchun Col(A) = span{(1, 2, 0)} — R^3 dagi to‘g‘ri chiziq. $(3,6,0)=3(1,2,0)$ ekanligidan, vektor shu to‘g‘ri chiziqda joylashgan: ha (masalan, $A\cdot\begin{bmatrix}3\\0\end{bmatrix}=(3,\,6,\,0)$).

Masala 20. Transponir va 3×3 **determinant.** $A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 0 & 0 & 2\end{bmatrix}$ **uchun** $A^Tva\det(A)$ **ni toping.**

Yechim. $A^T=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 0 & 1 & 2\end{bmatrix}$. A yuqori uchburchakli, shuning uchun $\det A=1\cdot 1\cdot 2=2$ (diagonal elementlar ko‘paytmasi); transponir determinantni o‘zgartirmaydi: $\det(A^T)=2$.

[Manba: 7-hafta Mustaqil o‘qitish topshiriqlari, Chiziqli algebra $I(Kuz2026)$ — 50 ta masala yechimsiz; yechimlar python tili

yordamida tuzilgan va tekshirilgan]

Mustaqil o‘qitish topshiriqlari — baholanmaydigan mustaqil amaliyot (ko‘rsatmalar: vektor-ustunlar, oraliq bosqichlar, javoblar asosida kichik fazolar). Fayl qismlari: $A(1-12)$ — transponiranіye, $B(13-20)$ — almashtirishlar va pivoting, $C(21-32)$ — kichik fazolar mezonlari, $D(33-42)$ — chiziqli qobiqlar va matritsa kichik fazolari, $E(43-46)$ — aralash qo‘llanilishi, $F(47-50)$ — murakkab masalalar. Mashqlar 1–20 auditoriya masalalari 1–20 bilan bir xil (pastda — ixcham yechimlar, to‘liqlari yuqorida); qolgan 30 tasi to‘liq yechilgan.

Mashq 1. $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ **uchun** A^T **ni hisoblang.**

Yechim. $A^T=\begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}$ (satrlar ustunlarga aylandi); batafsil — yuqoridagi 1-masala.

Mashq 2. $A=\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}$, $B=\begin{bmatrix}2 & 1\\ 0 & -1\end{bmatrix}$ **uchun** $(A+B)^T$ **ni hisoblang.**

Yechim. $A+B=\begin{bmatrix}3 & 1\\ 2 & 2\end{bmatrix}$, shuning uchun $(A+B)^T=\begin{bmatrix}3 & 2\\ 1 & 2\end{bmatrix}=A^T+B^T$.

Mashq 3. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ **uchun** $A^T A$ **ni toping.**

Yechim. $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ — natija simmetrik.

Mashq 4. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ **uchun** $(AB)^T$ **ni toping.**

Yechim. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, demak $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix} = B^T A^T$.

Mashq 5. $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ **matritsasi simmetrikmi?**

Yechim. Ha: $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$ — diagonal tashqari elementlar bir xil ($a_{12} = a_{21} = -3$).

Mashq 6. $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ **ni simmetrik va antisimmetrik qismlarga ajrating.**

Yechim. $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; tekshirish: $S + K = A$.

Mashq 7. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** PA **ni toping.**

Yechim. $PA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ — chapdan ko‘paytirish $1va2$-qatorlarni almashtirdi.

Mashq 8. $AvaP$ **uchun** AP **ni toping** ($A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$)**.**

Yechim. $AP = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ — o‘ngdan ko‘paytirish $1va2$-ustunlarni almashtirdi; shu bilan birga $AP \neq PA$.

Mashq 9. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **uchun** P^{-1} **ni toping.**

Yechim. $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; tekshirish: $P^T P = I$.

Mashq 10. 3×3 **matritsasi** $1va3$**-qatorlarni almashtiruvchi** P **almashtirish matritsasini toping.**

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — $I_3 da 1 va 3$-qatorlarni almashtirish; PA har qanday matritsaning birinchi va uchinchi qatorlarini almashtiradi.

Mashq 11. $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ **uchun** P, L, U **ni toping.**

Yechim. Nol bosh element almashtirishni talab qiladi: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; allaqachon yakunlangan, shuning uchun $L = I_2, U = PA$; natija: $PA = LU$.

Mashq 12. $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ **uchun qatorlarni almashtirmasdan** L, U **ni toping.**

Yechim. $m_{21} = \frac{6}{2} = 3$; $R2 \leftarrow R2 - 3R1$ bosqichi $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ ni beradi; $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$; tekshirish: $LU = A$.

Mashq 13. $W =$ **{**(x, y)**:** $y =$ **3x}** R^2 **ning kichik fazomiy?**

Yechim. Ha: $W = \{t(1, 3)$: $t \in R\}$, nol vektor $t = 0 da$, va $a(s,$ 3s$) + b(t,$ 3t$) = (as + bt, 3(as + bt)) \in W$.

Mashq 14. $S =$ **{**(x, y)**:** $y =$ **3x** $+2$**} kichik fazomiy?**

Yechim. Yo‘q: (0, 0) tenglamani qanoatlantirmaydi $(0 \neq 3 \cdot 0 + 2)$ — nol vektor yo‘q.

Mashq 15. $W =$ **{**(x, y, z)**:** $x + y - z = 0$**} kichik fazomiy?**

Yechim. Ha: $f(x, y, z) = x + y - z$ chiziqli, $f(0,\ 0,\ 0) = 0 va f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0 u,\ v \in W$ uchun.

Mashq 16. $(4,\ 1) ni (1,\ 1) va (1,\ -1)$ **vektorlari kombinatsiyasi sifatida ifodalang.**

Yechim. $c_1 + c_2 = 4$, $c_1 - c_2 = 1 \to c_1 = \frac{5}{2}$, $c_2 = \frac{3}{2}$; tekshirish: $(\frac{5}{2})(1,\ 1) + (\frac{3}{2})(1,\ -1) = (4,\ 1)$.

Mashq 17. $(1,\ 2,\ 0) va (2,\ 4,\ 0)$ **vektorlarining chiziqli qobig‘ini toping.**

Yechim. $(2, 4, 0) = 2(1, 2, 0)$, shuning uchun span{(1, 2, 0), (2, 4, 0)} = $\{t(1, 2, 0)$: $t \in R\}$ — to‘g‘ri chiziq.

Mashq 18. $A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$ **uchun Null**(A) **ni toping.**

Yechim. $x_1 + 2x_2 - x_3 = 0 \to x_1 = -2$s $+t, x_2 = s, x_3 = t$; Null(A) = span{(−2, 1, 0), (1, 0, 1)}.

Mashq 19. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$ **uchun Col**(A) **ni toping; (3, 6, 0) unga tegishlimi?**

Yechim. Ustunlar proporsional: Col(A) = span{(1, 2, 0)}; $(3, 6, 0) = 3(1, 2, 0)$ bo‘lgani uchun, vektor Col(A) ga tegishli.

Mashq 20. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ **uchun** $A^T va \det(A)$ **ni toping.**

Yechim. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$; A yuqori uchburchakli, shuning uchun $\det A = 1 \cdot 1 \cdot 2 = 2 = \det(A^T)$.

Mashq 21. O‘lchamlar. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ **uchun** A^T **ni hisoblang va o‘lchamlarni ko‘rsating.**

Yechim. A — 2×3; $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ — 3×2: o‘lchamlar almashtiriladi, $(m \times n)^T = n \times m$.

Mashq 22. Isbot (yig‘indi). $(A+B)^T = A^T + B^T$ **ni element bo‘yicha isbotlang.**

Yechim. $A = (a_{ij}), B = (b_{ij}) m \times n$ o‘lchamli va har bir i, j uchun: $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$. Elementlar bir xil — matritsalar teng.

Mashq 23. Isbot (ko‘paytma). $(AB)^T = B^T A^T$ **tenglamani mos matritsalar uchun isbotlang.**

Yechim. A — $m \times n, B$ — $n \times p$ bo‘lsin. Har bir i, j uchun: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Demak, $(AB)^T = B^T A^T$: ko‘paytuvchilar tartibi o‘zgaradi.

Mashq 24. 3×2 **uchun** $A^T A$ **ko‘paytmasi.** $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix}$ **uchun** $A^T A$ **ni hisoblang (faylda matritsa** 3×2 **qator bo‘yicha yozilgan).**

Yechim. $A^T A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 10 \end{bmatrix}$ — natija simmetrik, nazariya bashorat qilgandek (mashq 26).

Mashq 25. Simmetrik va antimetrik qismlar. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$ **uchun simmetrik va antimetrik qismlarini toping.**

Yechim. $A^T = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$. $S = (A + A^T)/2 = [1\frac{1}{2};\ \frac{1}{2}4]$, $K = (A - A^T)/2 = [0\frac{5}{2};\ -5/2\ 0]$. Tekshirish: $S + K = A, S^T = S, K^T = -K$.

Mashq 26. $A^T A$ **simmetrikligi. Ixtiyoriy haqiqiy matritsa** A **uchun** $A^T A$ **simmetrik ekanligini isbotlang.**

Yechim. $(A^T A)^T = A^T (A^T)^T = A^T A ((AB)^T = B^T A^T$ qoidasi): matritsa o‘ziga transponir bo‘lgani uchun simmetrikdir ($Am \times n$ o‘lchamli bo‘lganda $A^T A$ ko‘paytmasi $n \times n$ kvadrat matritsadir).

Mashq 27. Skalyar va tashqi ko‘paytma. $u = (2, -1)^T$, $v = (3, 4)^T$ **uchun** $u^T vvauv^T$ **ni hisoblang.**

Yechim. $u^T v = 2 \cdot 3 + (-1) \cdot 4 = 2$ — son (skalyar ko‘paytma); uv$^T = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 4 \end{bmatrix} =$

$\begin{bmatrix} 6 & 8 \\ -3 & -4 \end{bmatrix}$ — 2×2 matritsa (tashqi ko‘paytma).

Mashq 28. $1va2$**-qatorlarni almashtirish.** I_3 **matritsasining** $1va2$**-qatorlarini almashtiruvchi permutatsiya matritsasini tuzing.**

Yechim. $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ — I_3 matritsasi $1va2$-qatorlari almashtirilgan. Har qanday uch qatorli A uchun PA birinchi ikki qatorni almashtiradi: $P \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$.

Mashq 29. $PAvaAP$ **solishtirish.** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $vaA = \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix}$ **uchun** $PAvaAP$ **ni solishtiring.**

Yechim. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 2 & 5 \end{bmatrix}$ — qatorlar almashtirilgan; $AP = \begin{bmatrix} 2 & 5 \\ 7 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 1 & 7 \end{bmatrix}$ — ustunlar almashtirilgan. $PA \neq AP$: chap ko‘paytirish qatorlarga, o‘ng ko‘paytirish esa ustunlarga ta’sir qiladi.

Mashq 30. $P^TP = I$ **tekshiruvi.** $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ **uchun** $P^TP = I$ **ekanligini tekshiring.**

Yechim. $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; $P^TP = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = I$. P^T qatorlari — P ustunlari (e_2, e_3, e_1), ularning skalyar ko‘paytmalari δ_{ij} ni beradi; natijada: $P^{-1} = P^T$.

Mashq 31. Permutatsiya matritsasining determinantini topish. $1va3$**-qatorlarni almashtiruvchi permutatsiya matritsasining determinantini toping.**

Yechim. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; bitta qatorlarni almashtirish belgini o‘zgartiradi: $\det P = -1$.

Birinchi qator bo‘yicha yoyish orqali tekshirish: $1 \cdot (-1)^{1+3} \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 1 \cdot 1 \cdot (-1) = -1$.

Mashq 32. 2×2 **uchun** $PA = LU$**.** $A = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}$ **uchun** P, L, U **ni toping.**

Yechim. Bosh element $a_{11} = 0$ — almashtirish kerak: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$. Bosh

element 4 ostida $0 bor$, chiqarish tugallandi: $L = I_2$, $U = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$; natija: $PA = LU$.

Mashq 33. Qisman bosh elementni tanlash. Qisman bosh elementni tanlash (partial pivoting) usulini $A = \begin{bmatrix} 0.001 & 1 \\ 2 & 1 \end{bmatrix}$ ga qo‘llang; almashtirish oldingi va keyingi ko‘paytuvchilarni solishtiring.

Yechim. Almashtirmasdan: bosh element 0.001, ko‘paytuvchi $m_{21} = \frac{2}{0}.001 = 2000$ — juda katta ($2000\cdot$(birinchi qator) ayirish xatolarni kuchaytirardi). Qisman tanlov birinchi ustundagi modul bo‘yicha eng kandidatni ($son 2$) oladi; almashtirgandan keyin ($P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 2 & 1 \\ 0.001 & 1 \end{bmatrix}$) ko‘paytuvchi $m_{21} = 0.\frac{001}{2} = 0.0005$. Almashtirish ko‘paytuvchini $2000 dan 0.0005$ gacha kamaytirdi ($4 \cdot 10^6$ marta).

Mashq 34. Altbo‘shmi? $x+$ 2y $= 0$. $W =$ {(x, y): $x+$ 2y $= 0$} to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. Ha. $(1) 0 + 2 \cdot 0 = 0$ — nol vektor W da. $(2) x+$ 2y $= 0 va x' +$ 2y$' = 0 dan (x + x') + 2(y + y') = 0$ kelib chiqadi. $(3) (cx) + 2(cy) = c(x + 2y) = 0$. Natija: $W =$ span{(−2, 1)} — koordinata boshidan o‘tgan to‘g‘ri chiziq.

Mashq 35. Altbo‘shmi? $x+$ 2y $= 3$. $W =$ {(x, y): $x+$ 2y $= 3$} to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. Yo‘q: $0 + 2 \cdot 0 = 0 \neq 3$ — nol vektor W da yo‘q, altbo‘shlik 0 ni o‘z ichiga olishi kerak (siljigan to‘g‘ri chiziq koordinata boshidan o‘tmaydi).

Mashq 36. Altbo‘shmi? $x \geq 0$. $W =$ {(x, y): $x \geq 0$} to‘plamining altbo‘shlik ekanligini tekshiring.

Yechim. Yo‘q. Nol vektor W da joylashgan va W qo‘shish nisbatan yopiq (manfiy bo‘lmagan abssissalar yig‘indisi manfiy bo‘lmagan). Ammo skalyarlar nisbatidan yopiqlik buzilgan: $(1, 0) \in W$, lekin $(-1) \cdot (1, 0) = (-1, 0) \notin W$. Yarim tekislik — altbo‘shlik emas.

Mashq 37. Untor? $x - y + z = 0$. Tekshirish kerakki, $W =$ {(x, y, z): $x - y + z = 0$} untor bo‘lishini.

Yechim. Ha: $f(x, y, z) = x - y + z$ chiziqli, $f(0, 0, 0) = 0 va f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0 u$, $v \in W$ uchun. Aniq: $x = y - z (x, y, z) = y(1, 1, 0) + z(-1, 0, 1)$ beradi, ya’ni $W =$ span{(1, 1, 0), (−1, 0, 1)} — koordinata boshidan o‘tgan tekislik.

Mashq 38. Chiziqli birikma. $(6,\ 2) ni (1,\ 1) va (1,\ -1)$ vektorlarining birikmasi sifatida ifodalash.

Yechim. $c_1 + c_2 = 6$, $c_1 - c_2 = 2 \to c_1 = 4$, $c_2 = 2$. Tekshirish: $4(1, 1) + 2(1, -1) = (6, 2)$.

Mashq 39. Qobiq geometriyasi. span**{(1, 2, 0), (0, 1, 2)} ni geometrik tasvirlang.**

Yechim. Vektorlar proporsional emas — qobiq ikki o‘lchovli: koordinata boshidan o‘tgan tekislik. Uning nuqtalari: $s(1, 2, 0) + t(0, 1, 2) =$ (s, 2s $+t$, 2t); parametrlarni chiqarib tashlab ($s = x, t = z/2$), 4x − 2y $+z = 0$ hosil bo‘ladi. Tekshirish: $4 \cdot 1 - 2 \cdot 2 + 0 = 0 va 4 \cdot 0 - 2 \cdot 1 + 2 = 0$.

Mashq 40. Nol-untor 2×3. $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (2×3 matritsa, qator bo‘yicha yozilgan) uchun Null(A) ni toping.

Yechim. 2-qatoridan: $x_2 = -x_3$; 1-qatoridan: $x_1 = -x_2 - x_3 = 0$. Erkin o‘zgaruvchi $x_3 = t$: $x = t(0, -1, 1)$. Natija: Null(A) = span{(0, −1, 1)} — R^3 dagi to‘g‘ri chiziq. Tekshirish: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Mashq 41. Ustunlar bo‘shlig‘i 3×2. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$ uchun Col(A) ni toping.

Yechim. Ustunlar $a_1 = (1, 2, 3) va a_2 = (2, 4, 6) = 2a_1$ proporsional, shuning uchun har qanday birikma a_1 ning karrali: Col(A) = span{(1, 2, 3)} — R^3 dagi to‘g‘ri chiziq ($(t$, 2t, 3t) vektorlari).

Mashq 42. Ustunlar bo‘shlig‘iga tegishlilik. (1, 1, 1) vektori $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ (3×2 matritsa) matritsasi uchun Col(A) ustunlar bo‘shligiga tegishlimi?

Yechim. $c_1(1, 0, 1) + c_2(0, 1, 1) = (1, 1, 1)$ ni qidiraylik: birinchi ikki koordinata $c_1 = c_2 = 1$ beradi, lekin uchinchisi $2 \neq 1$ — ziddiyat. Natija: $(1, 1, 1) \notin$ Col$(A) (Col(A)$—$z = x + y$ tekisligi, $1 \neq 1 + 1$).

Mashq 43. Untorlarning kesishishi. Ikkita untorning kesishmasi untor ekanligini isbotlang.

Yechim. $U, W \subseteq V$ untorlari uchun: $(1) 0 \in U \cap W$. (2) Agar $u, v \in U \cap W, u+v \in U va u+v \in W (har$ birining yopiq bo‘lishi), demak $u+v \in U \cap W$. (3) Agar $u \in U \cap W$, cu $\in U va cu \in W$, demak cu $\in U \cap W$. Test bajarildi.

Mashq 44. Untorlarning birlashmasi. Birlashmasi untor bo‘lmagan ikki untor keltiring.

Yechim. R^2 da o‘qlarni $U = \{(x, 0)\}$ va $W = \{(0, y)\}$ deb olaylik — ikkalasi ham untor. Birlashmada $(1, 0) va (0, 1) bor$, lekin ularning yig‘indisi (1, 1) hech qaysi o‘qda joylashmagan. Birlashma qo‘shishga nisbatan yopiq emas — untor emas.

Mashq 45. Simmetrik 3×3 matritsalar. 3×3 barcha haqiqiy matritsalar fazasining simmetrik 3×3 matritsalar to‘plami untormi?

Yechim. Ha. (1) Nol matritsa simmetrik. (2) Simmetrik S, T uchun $(S+T)^T = S^T + T^T = S + T$. $(3) (cS)^T = c \cdot S^T =$ cS. Yopiq bo‘lish bajarildi — untor.

Mashq 46. $p(0) = 1$ bo‘lgan ko‘phadlar. $p(0) = 1$ shartini qanoatlantiruvchi ko‘phadlar to‘plami untormi?

Yechim. Yo‘q. Nol ko‘phad $p(0) = 0 \neq 1$ beradi — nol vektori yo‘q. Bundan tashqari, $p(x) = 1 va q(x) = 1 + x$ uchun: $(p + q)(0) = 2 \neq 1$ — qo‘shishning yopiq bo‘lishi buzilgan. ($p(0) = 0$ bo‘lgan ko‘phadlar, aksincha, — untor.)

Mashq 47. Murakkab masala. Ixtiyoriy haqiqiy A matritsasi uchun Null$(A^T A)$ = Null(A) ni isbotlang.

Yechim. Ikkki kiritish. (1) Agar $Ax = 0, u$ holda $A^T Ax = A^T (Ax) = 0$, ya’ni Null$(A) \subseteq$ Null$(A^T A)$. $(2) A^T Ax = 0$ deb faraz qilaylik; chapdan x^T ko‘paytiramiz: $x^T A^T Ax = ||Ax||^2 = 0$. Ax koordinatalari kvadratlari yig‘indisi nolga teng, demak, $Ax = 0$. Natija:

Null(A^TA) = Null(A); xulosa — $AvaA^TAbirxil$ nol-untor va rangga ega (eng kichik kvadratlar fakti).

Mashq 48. Murakkab masala. P **— almashtiruvchi matritsa bo‘lsa, Col**$(PA) =$ P **Col**(A) **ni isbotlang.**

Yechim. Ta’rifiga ko‘ra Col(PA) = {PAx: $x \in R^n$ }, P Col(A) = {Pb: $b \in$ Col(A)}. (1) Agar $y =$ PAx, u holda $y = P(Ax)Ax \in$ Col(A) bilan — demak $y \in P$ Col(A). (2) Agar $y = Pbb = Ax \in$ Col(A) bilan, u holda $y =$ PAx $\in$ Col(PA). Natija: Col$(PA) = P$ Col(A) — qatorlarni almashtirish ustunlar bo‘shlig‘ining har bir nuqtasini bir xil almashtirish bilan o‘zgartiradi ($b \to Pb$ birma-bir, P teskari).

Mashq 49. Murakkab masala. Agar W **— untor bo‘lsa,** span$(W) = W$ **ni isbotlang.**

Yechim. span(W) — barcha chekli kombinatsiyalar $c_1w_1+...+c_kw_k$ to‘plami, bunda $w_i \in W$. (1) Har bir kombinatsiya W da joylashgan: $c_iw_i \in W$ (skalyarlar), W elementlari yig‘indisi Wda (qo‘shish). (2) Har bir $w \in W1 \cdot w$ kombinatsiyasidir. Natija: span$(W) = W$ — bu kichik fazo o‘zining barcha kombinatsiyalarini o‘z ichiga oladi.

Mashq 50. Murakkab masala. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ **uchun Col**(A) **va Null**(A) **ni tasvirlang.**

Yechim. Ustunlar $a_1 = (1,\ 0,\ 0)$, $a_2 = (2,\ 1,\ 0)$, $a_3 = (0,\ 1,\ 0) = a_2 - 2a_1$; uchinchi qator nolli, shuning uchun barcha ustunlar $z = 0$ tekisligida joylashgan, a_1, a_2 esa mustaqil. Natija: Col(A) = span{(1, 0, 0), (2, 1, 0)} = $\{(s + 2t, t, 0)\}$ — bu $z = 0$ tekisligi. Nol-fazo: 2-qatordan $x_2 = -x_3$, 1-qatordan $x_1 = -2x_2 = 2x_3$; $x_3 = t$: $x = t(2, -1, 1)$. Natija: Null(A) = span{(2, −1, 1)} — to‘g‘ri chiziq. Tekshirish: $A \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.